DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT

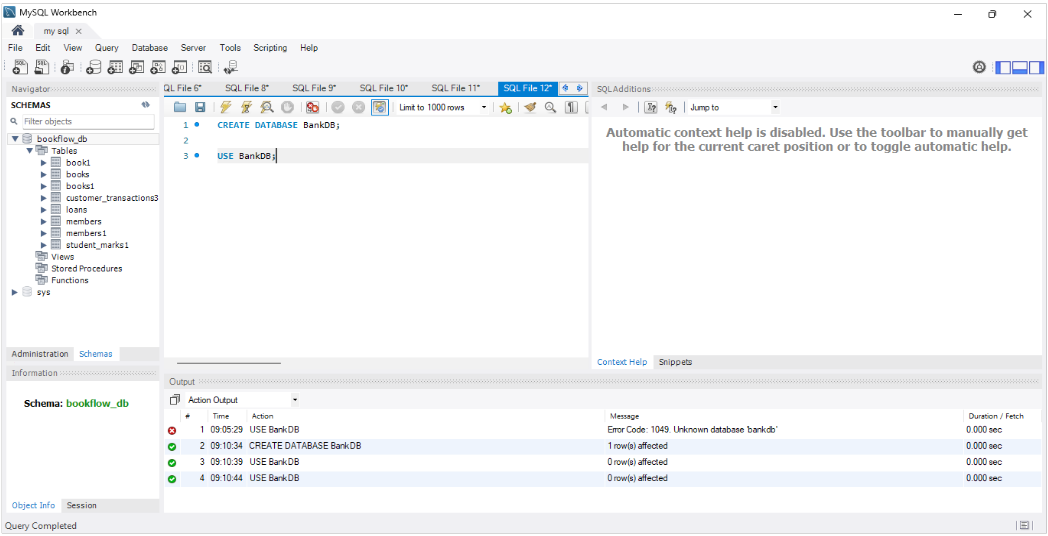
Sec:- 3

Name:- Srithika Kambhammettu

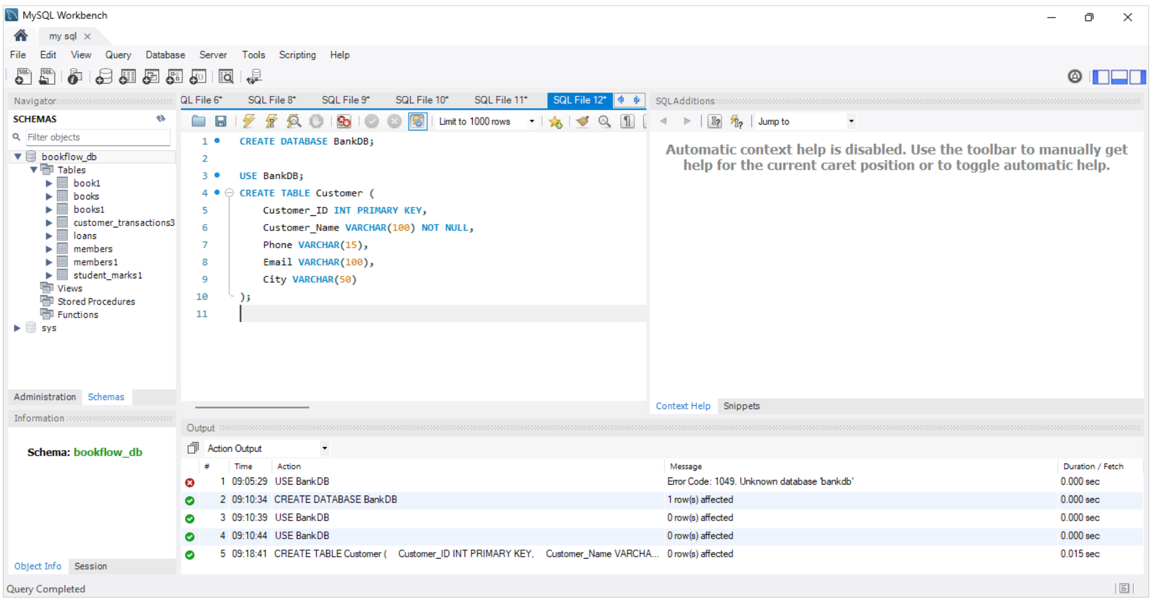
Roll no:- 2520030361

6 - Topic: Bank Database – Triggers and Stored Procedures

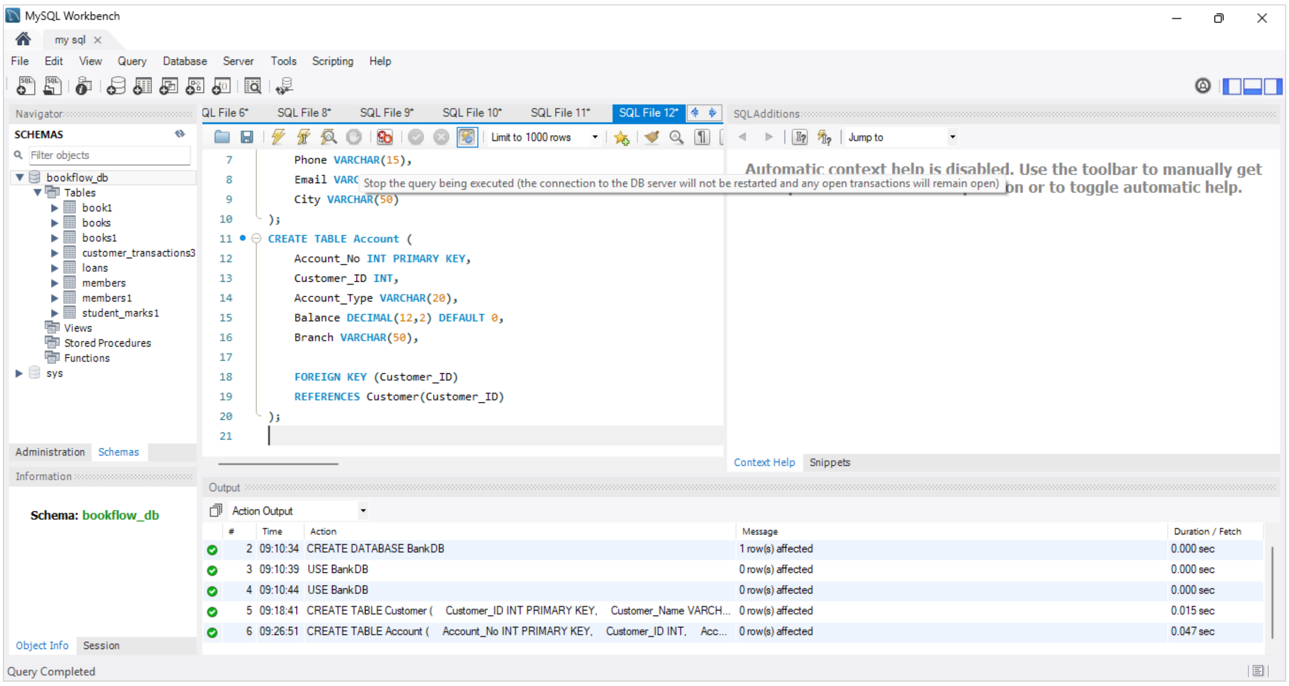
1. CREATE DATABASE



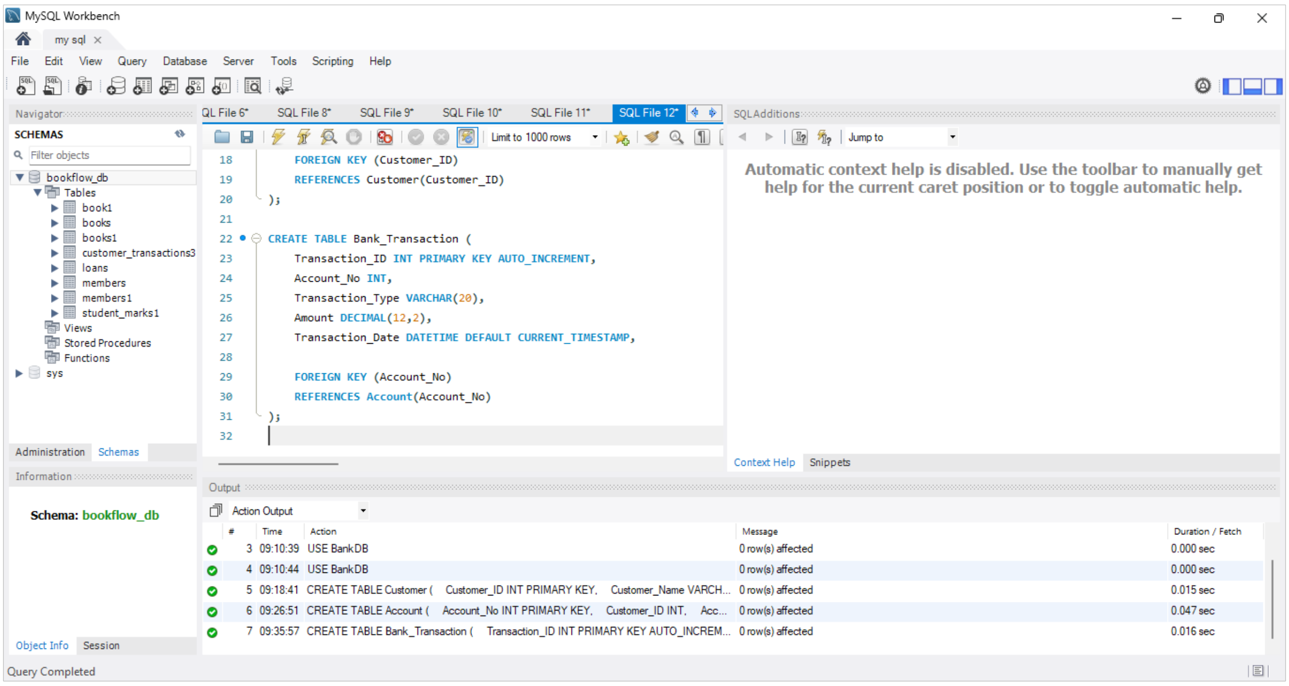
1. 2. CREATE CUSTOMER TABLE



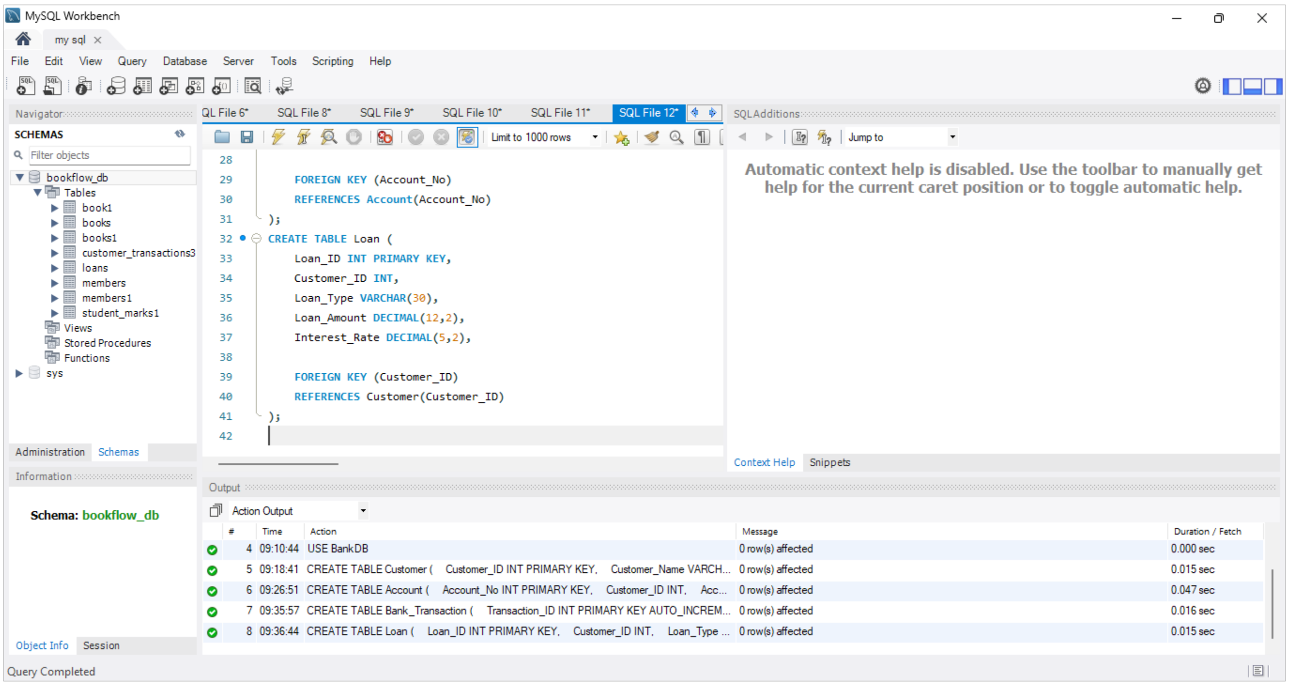
3. CREATE ACCOUNT TABLE



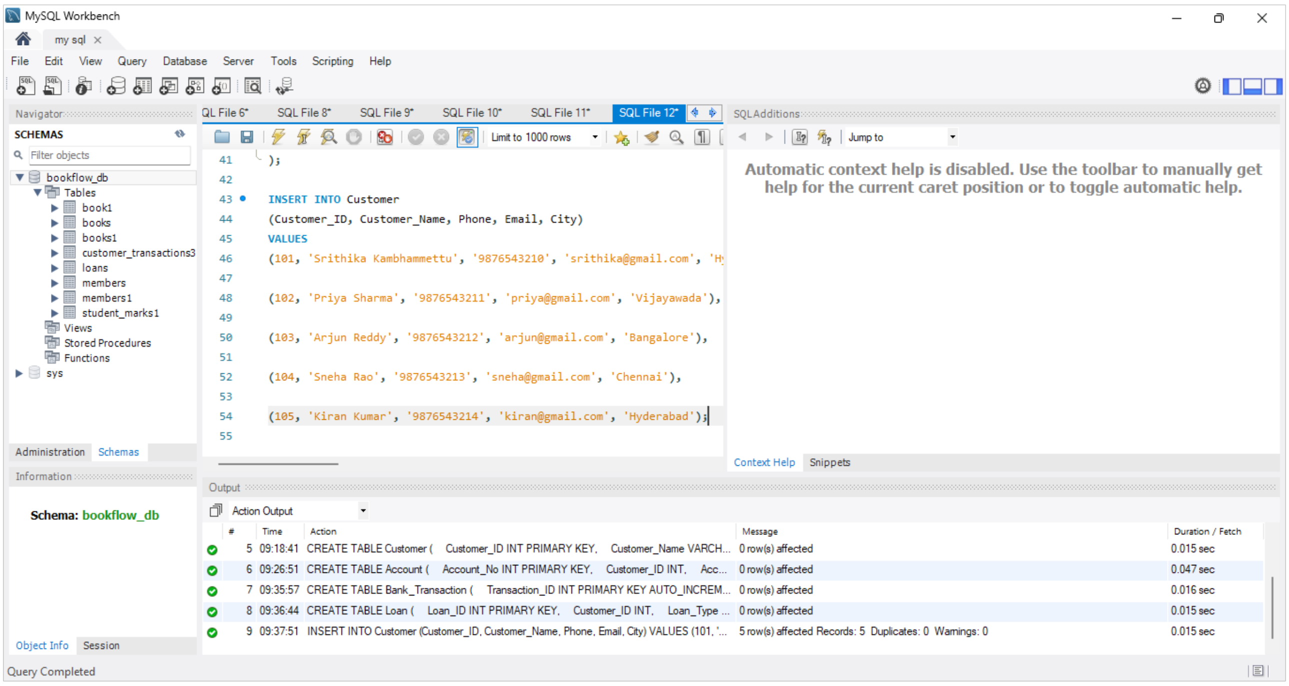
4. CREATE TRANSACTION TABLE



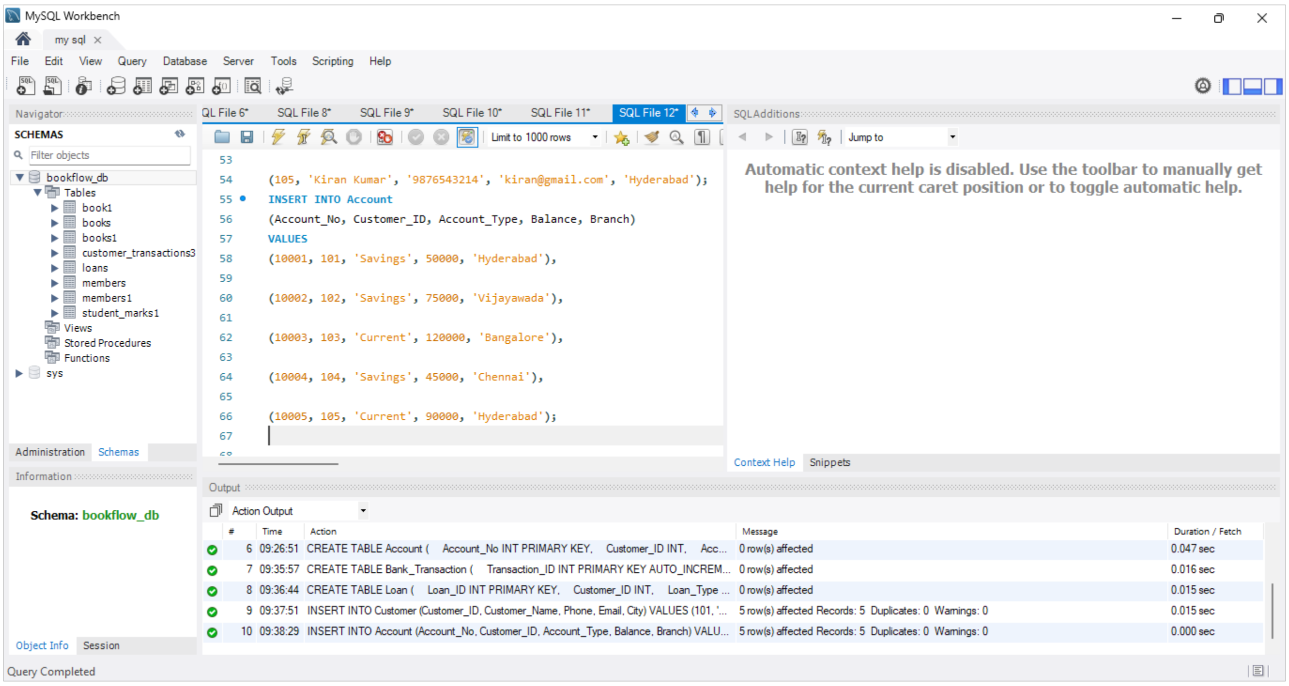
5. CREATE LOAN TABLE



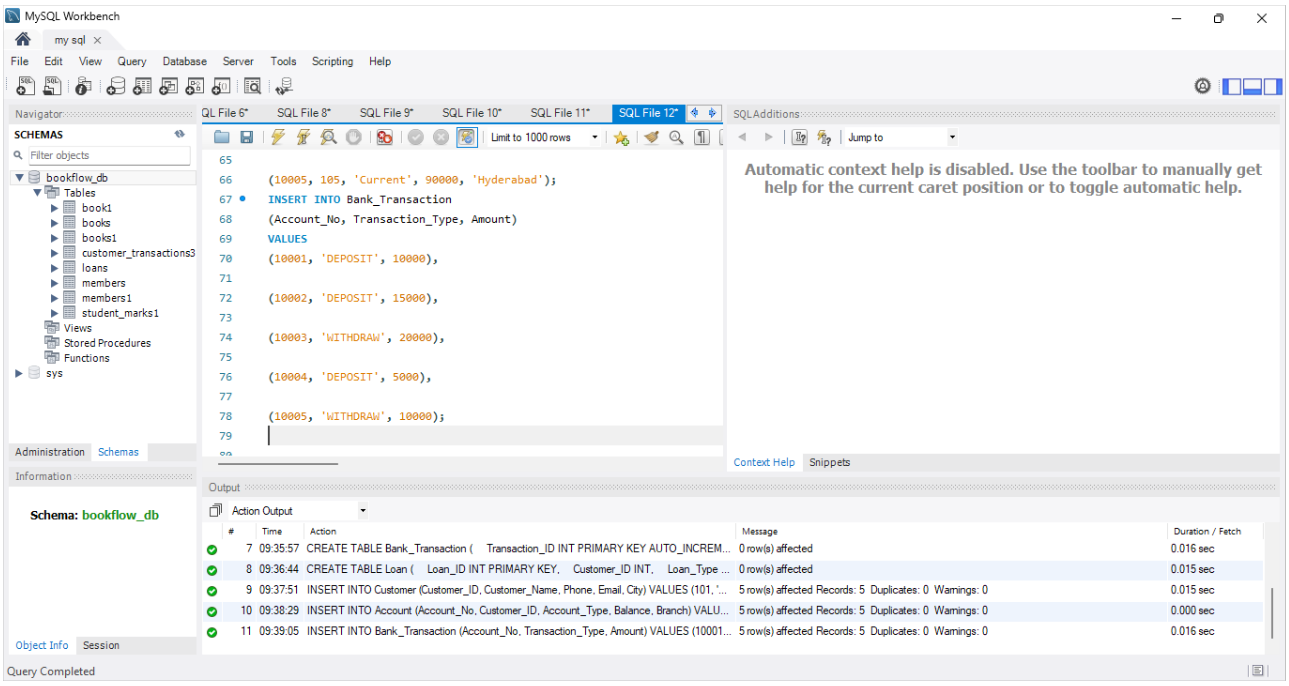
6. INSERT CUSTOMER DATA



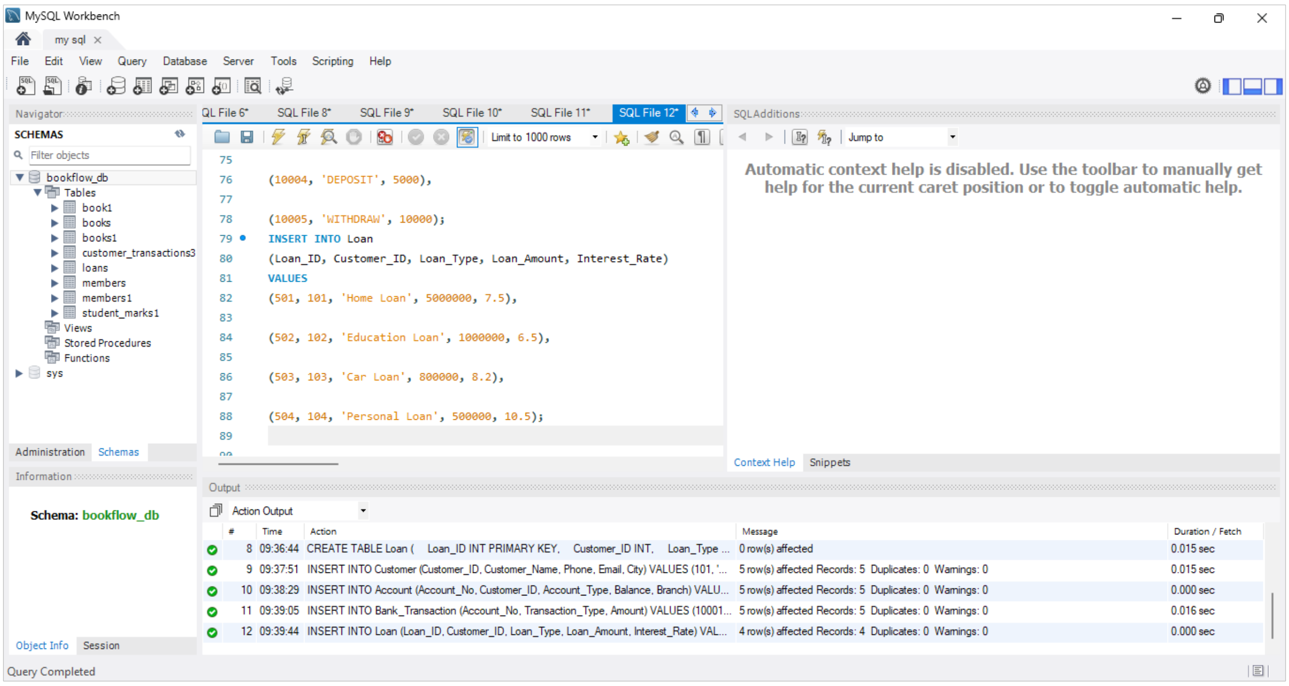
7. INSERT ACCOUNT DATA



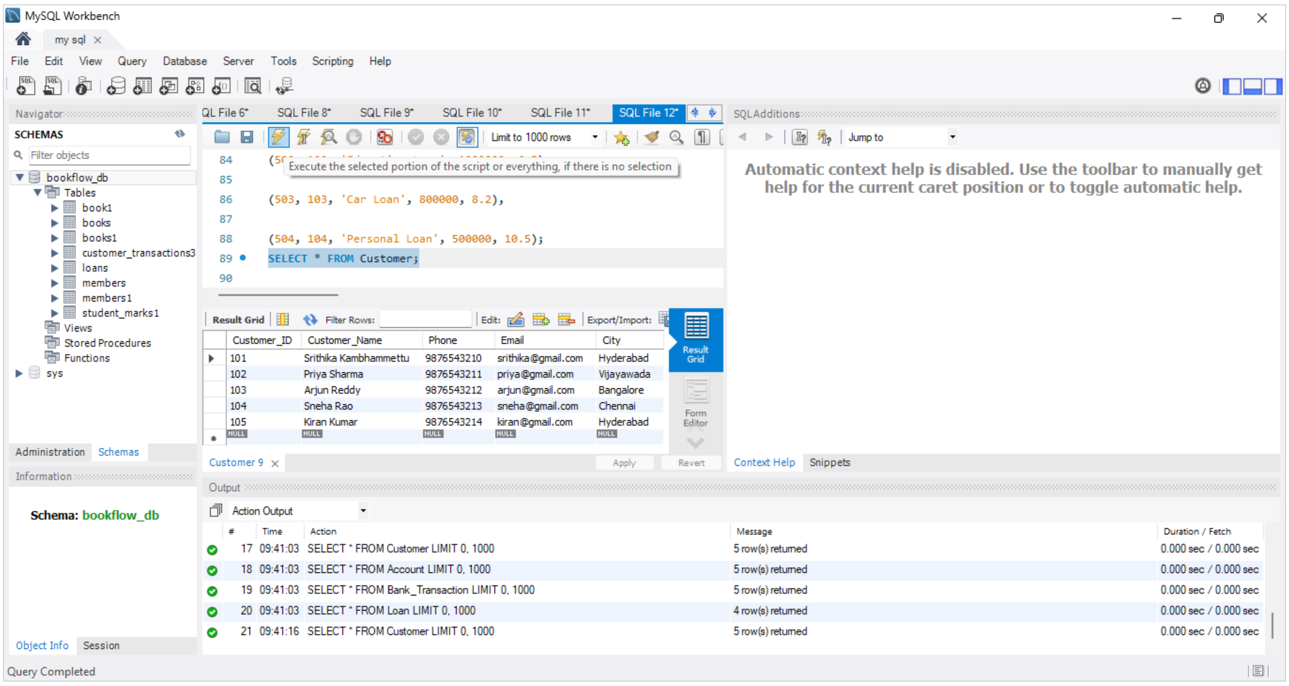
8. INSERT TRANSACTION DATA

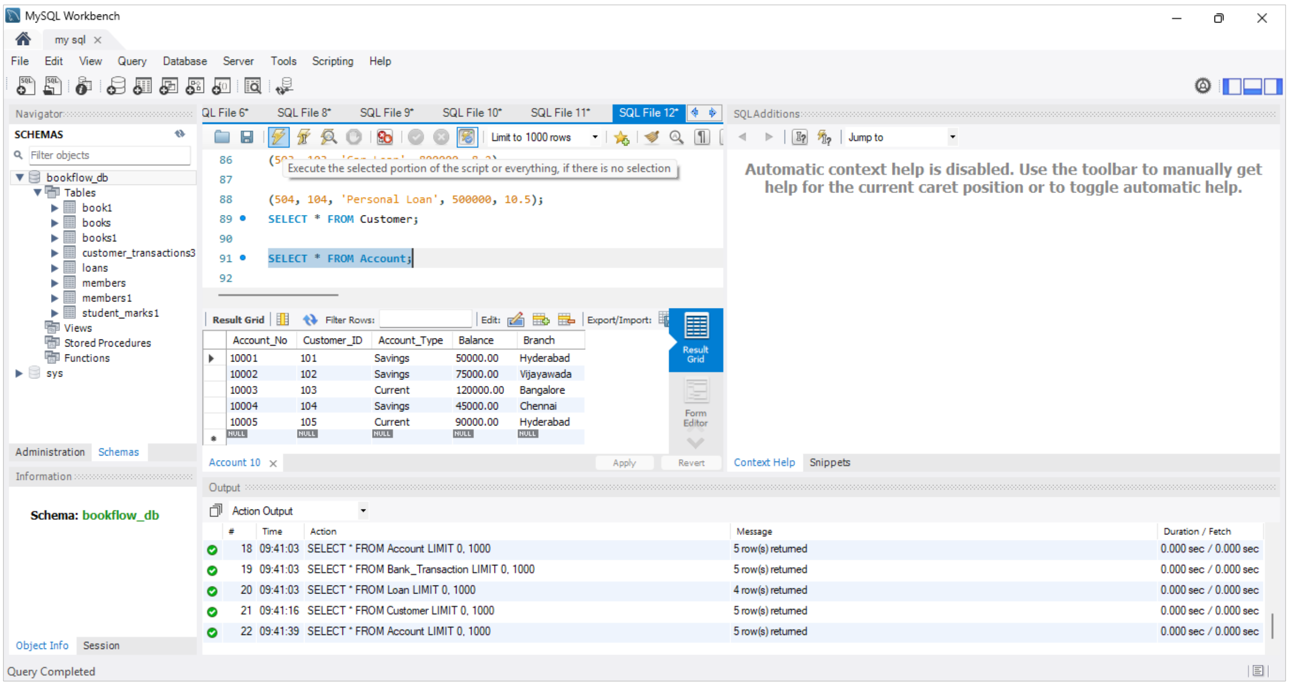


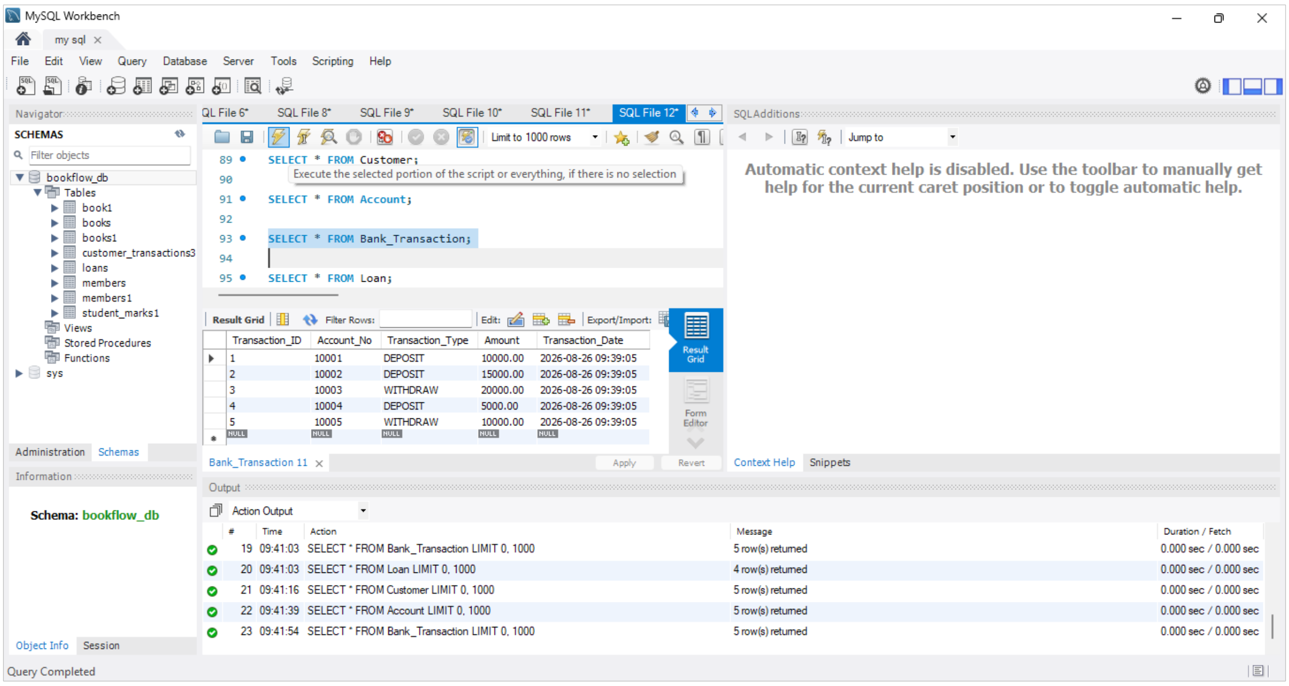
9. INSERT LOAN DATA

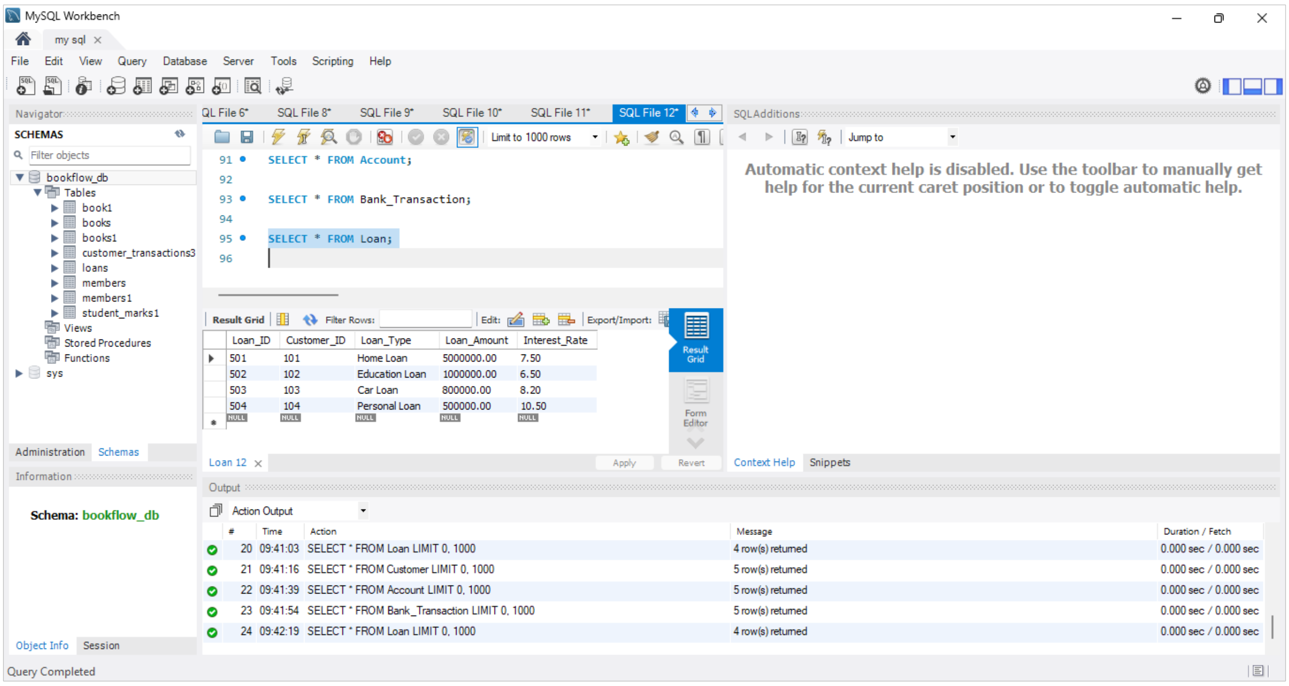


10. DISPLAY TABLE DATA

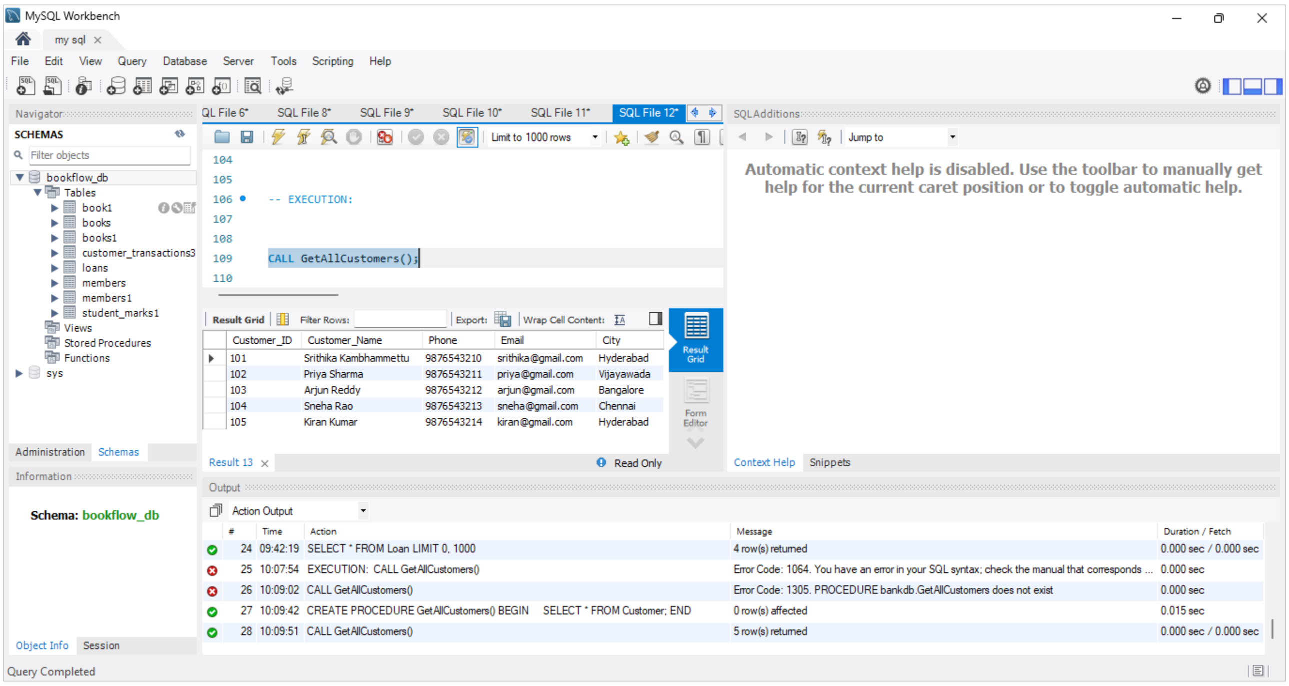



MySQL Workbench
(504, 104, 'Personal Loan', 500000, 10.5);
SELECT * FROM Customer;
SELECT * FROM Account;
Execute the selected portion of the script or everything, if there is no selection
Account_No Customer_ID Account_Type Balance Branch
10001 101 Savings 50000.00 Hyderabad
10002 102 Savings 75000.00 Vijayawada
10003 103 Current 120000.00 Bangalore
10004 104 Savings 45000.00 Chennai
10005 105 Current 90000.00 Hyderabad
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
Schema: bookflow_db
18 09:41:03 SELECT * FROM Account LIMIT 0, 1000 5 row(s) returned 0.000 sec / 0.000 sec
19 09:41:03 SELECT * FROM Bank_Transaction LIMIT 0, 1000 5 row(s) returned 0.000 sec / 0.000 sec
20 09:41:03 SELECT * FROM Loan LIMIT 0, 1000 4 row(s) returned 0.000 sec / 0.000 sec
21 09:41:16 SELECT * FROM Customer LIMIT 0, 1000 5 row(s) returned 0.000 sec / 0.000 sec
22 09:41:39 SELECT * FROM Account LIMIT 0, 1000 5 row(s) returned 0.000 sec / 0.000 sec
Query Completed


MySQL Workbench
SELECT * FROM Customer;
SELECT * FROM Account;
SELECT * FROM Bank_Transaction;
SELECT * FROM Loan;
Execute the selected portion of the script or everything, if there is no selection
Transaction_ID Account_No Transaction_Type Amount Transaction_Date
1 10001 DEPOSIT 10000.00 2026-08-26 09:39:05
2 10002 DEPOSIT 15000.00 2026-08-26 09:39:05
3 10003 WITHDRAW 20000.00 2026-08-26 09:39:05
4 10004 DEPOSIT 5000.00 2026-08-26 09:39:05
5 10005 WITHDRAW 10000.00 2026-08-26 09:39:05
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
Schema: bookflow_db
19 09:41:03 SELECT * FROM Bank_Transaction LIMIT 0, 1000 5 row(s) returned 0.000 sec / 0.000 sec
20 09:41:03 SELECT * FROM Loan LIMIT 0, 1000 4 row(s) returned 0.000 sec / 0.000 sec
21 09:41:16 SELECT * FROM Customer LIMIT 0, 1000 5 row(s) returned 0.000 sec / 0.000 sec
22 09:41:39 SELECT * FROM Account LIMIT 0, 1000 5 row(s) returned 0.000 sec / 0.000 sec
23 09:41:54 SELECT * FROM Bank_Transaction LIMIT 0, 1000 5 row(s) returned 0.000 sec / 0.000 sec
Query Completed

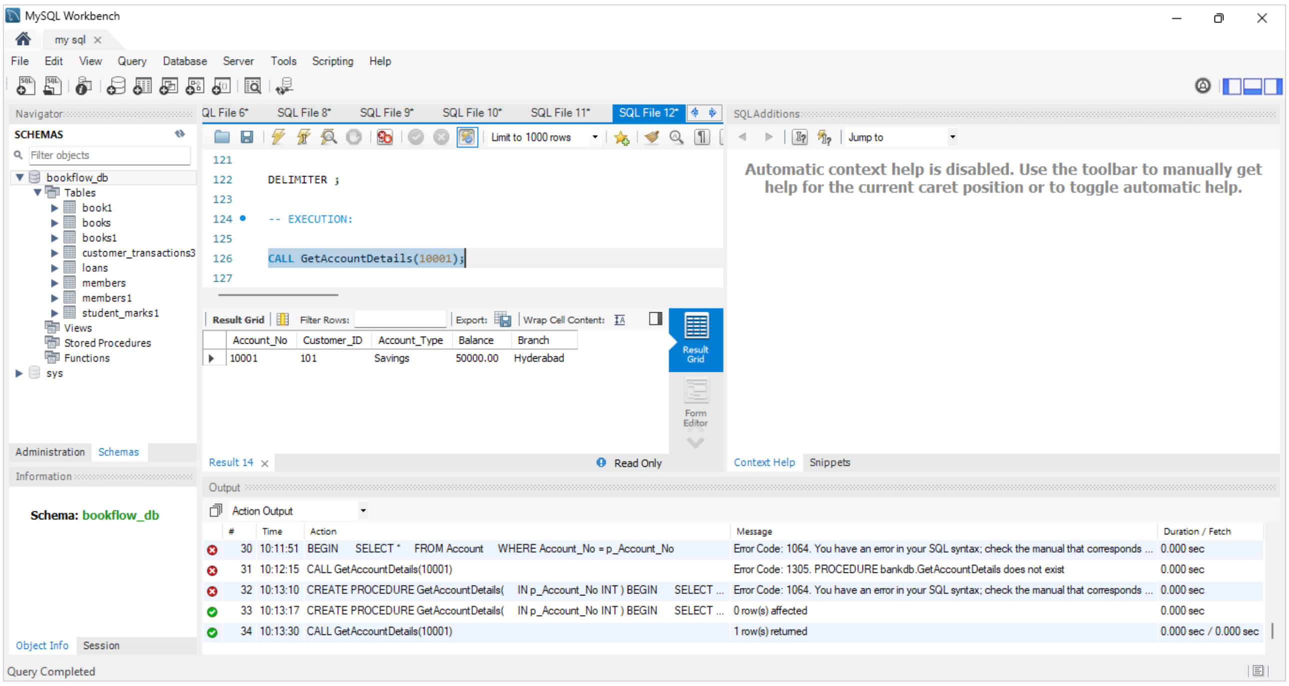


11. PROCEDURE TO DISPLAY ALL CUSTOMERS

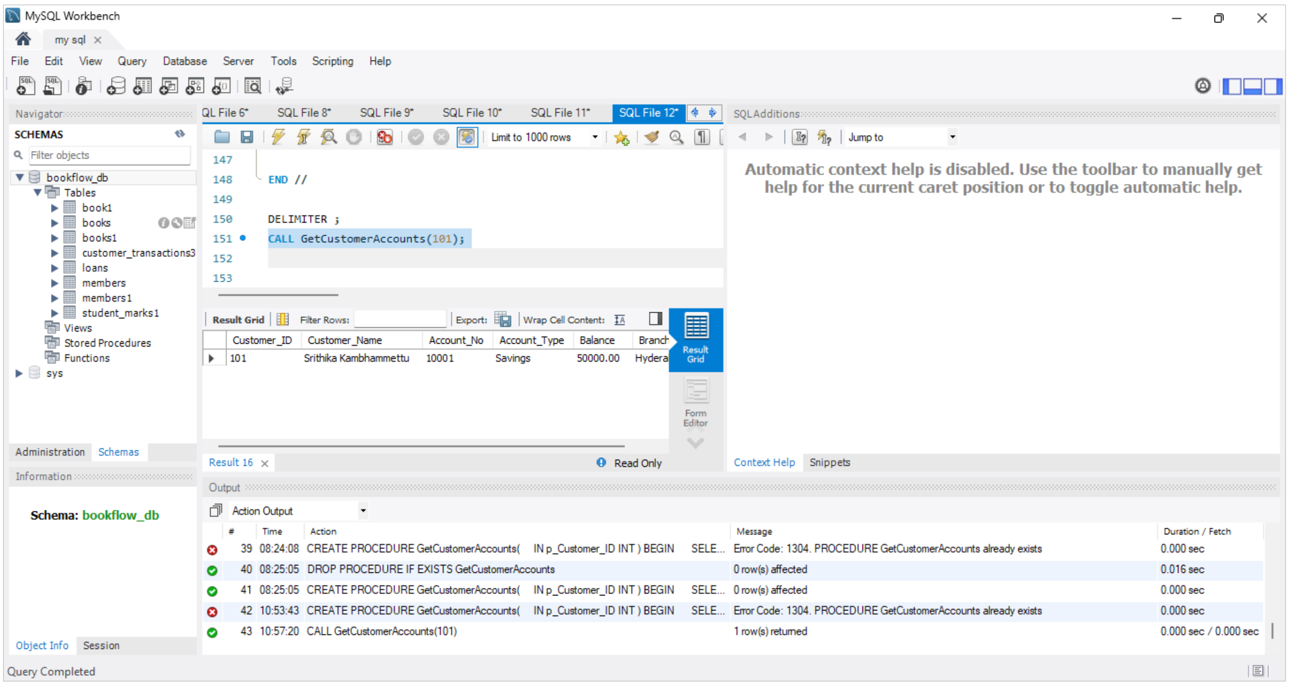


12. PROCEDURE TO DISPLAY ACCOUNT DETAILS

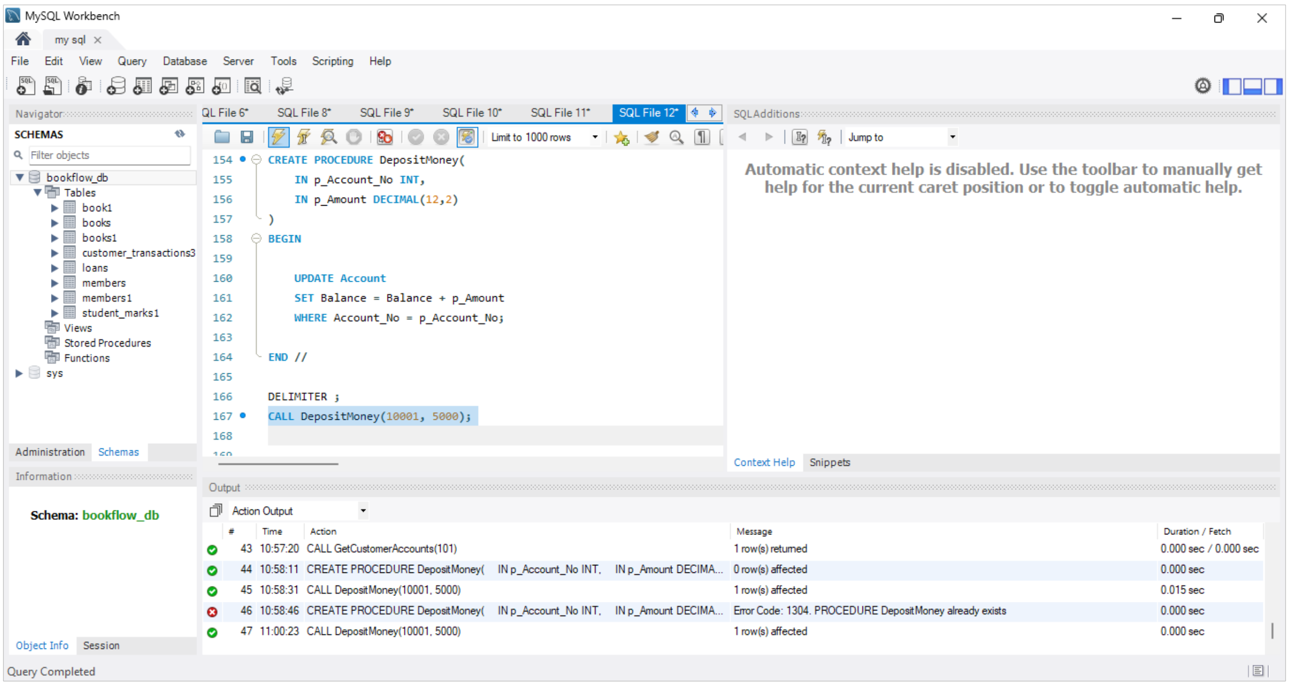
FOR A GIVEN ACCOUNT NUMBER



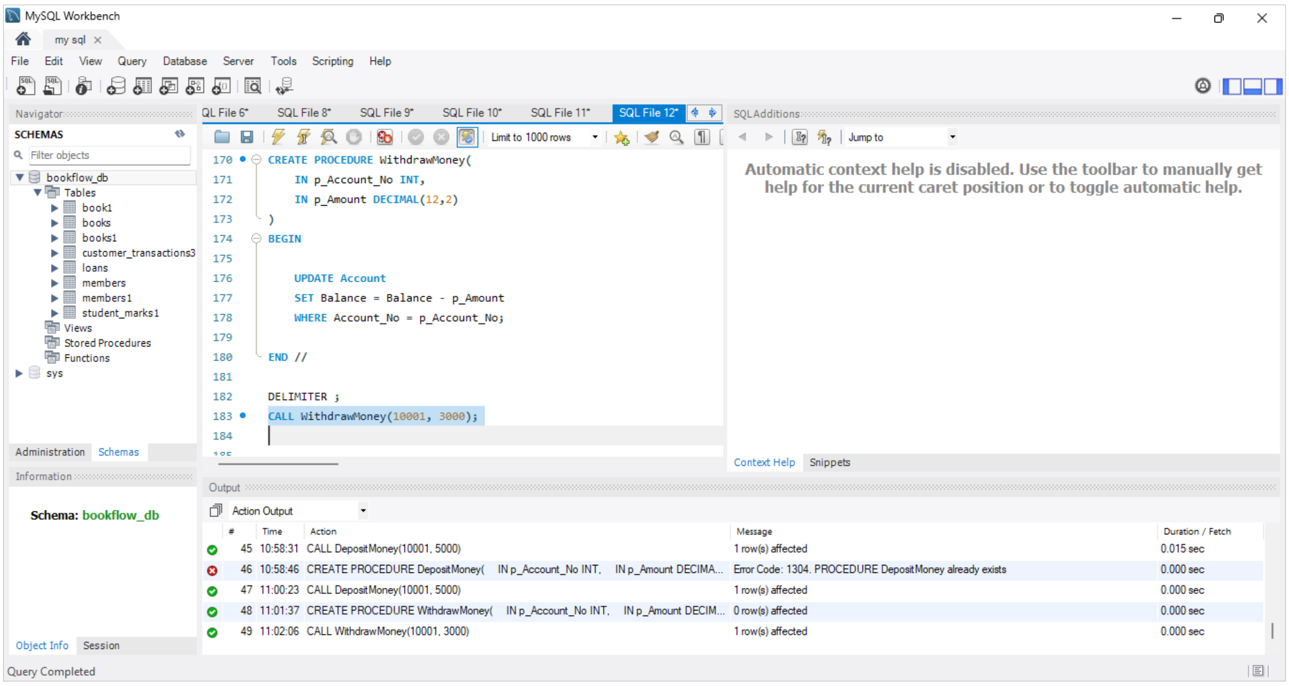
13. PROCEDURE TO FIND CUSTOMER ACCOUNTS



14. PROCEDURE TO DEPOSIT MONEY

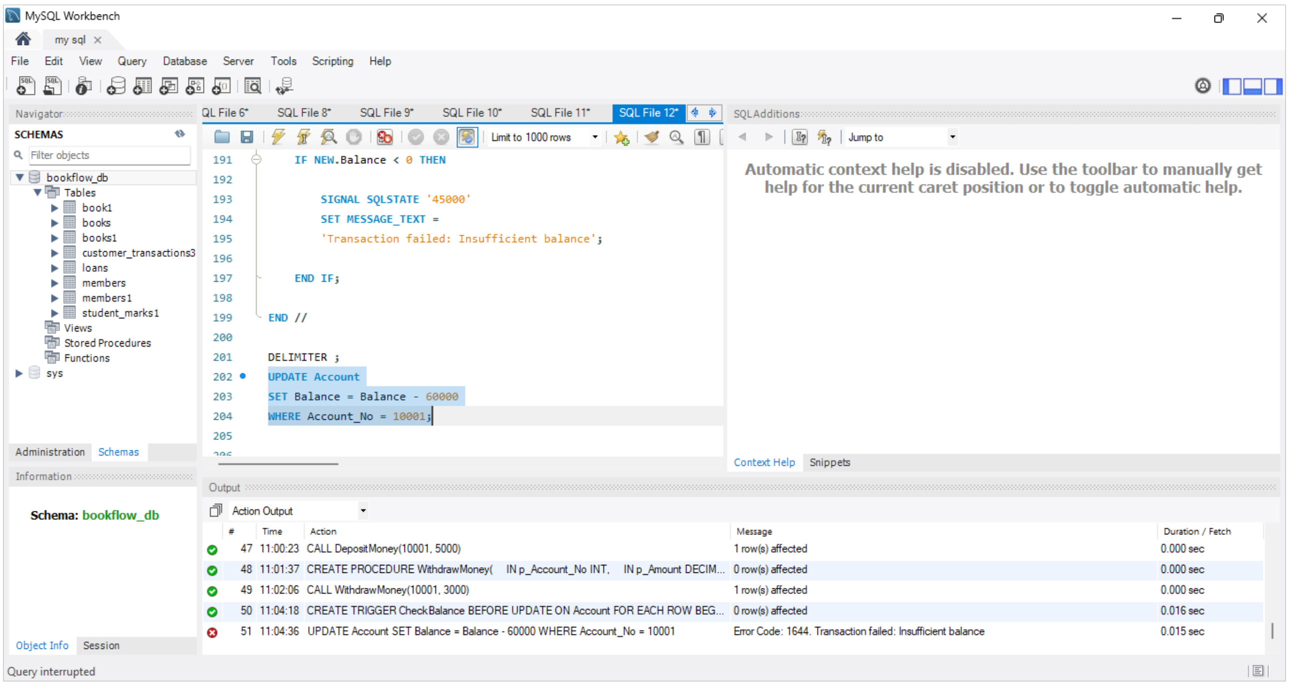


15. PROCEDURE TO WITHDRAW MONEY

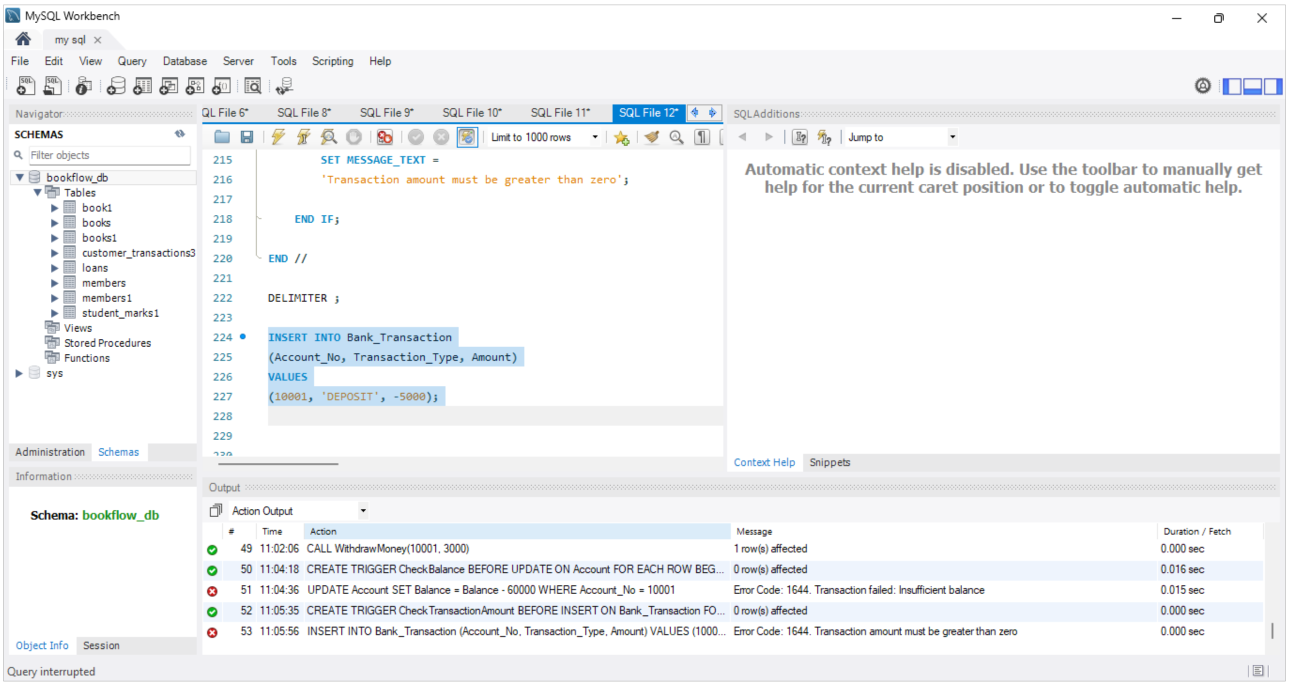


TRIGGERS

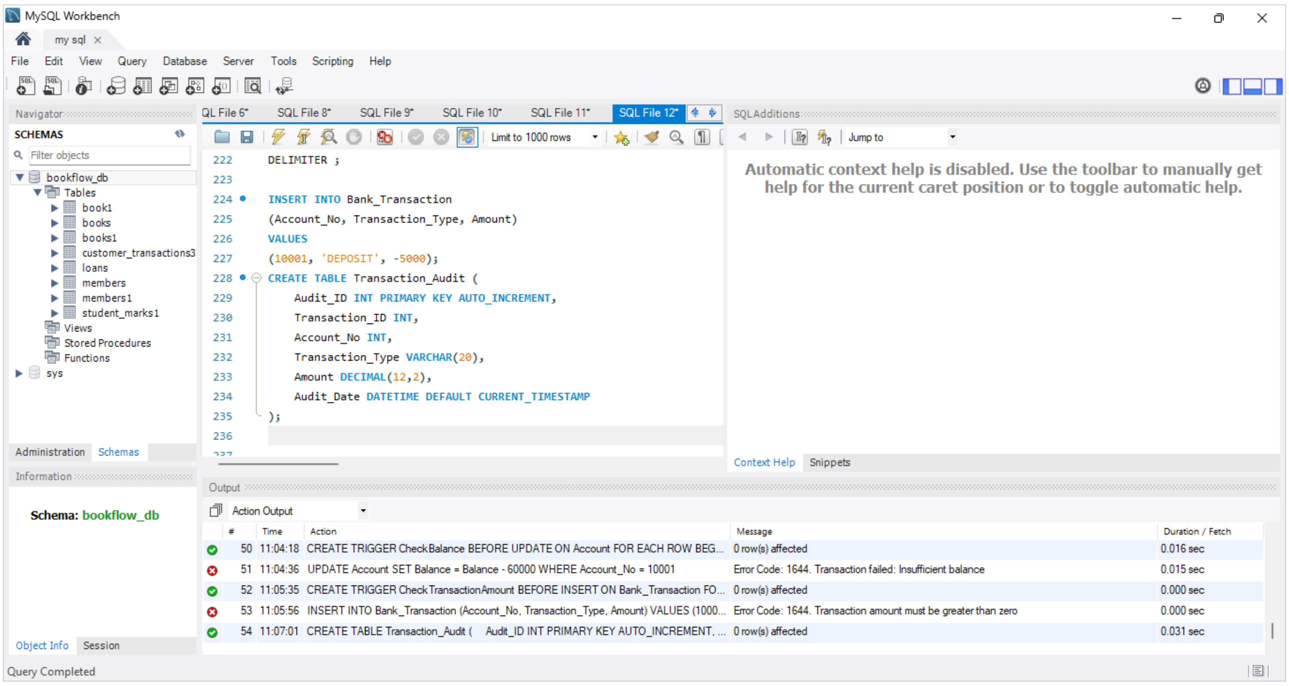
16. TRIGGER – PREVENT NEGATIVE BALANCE



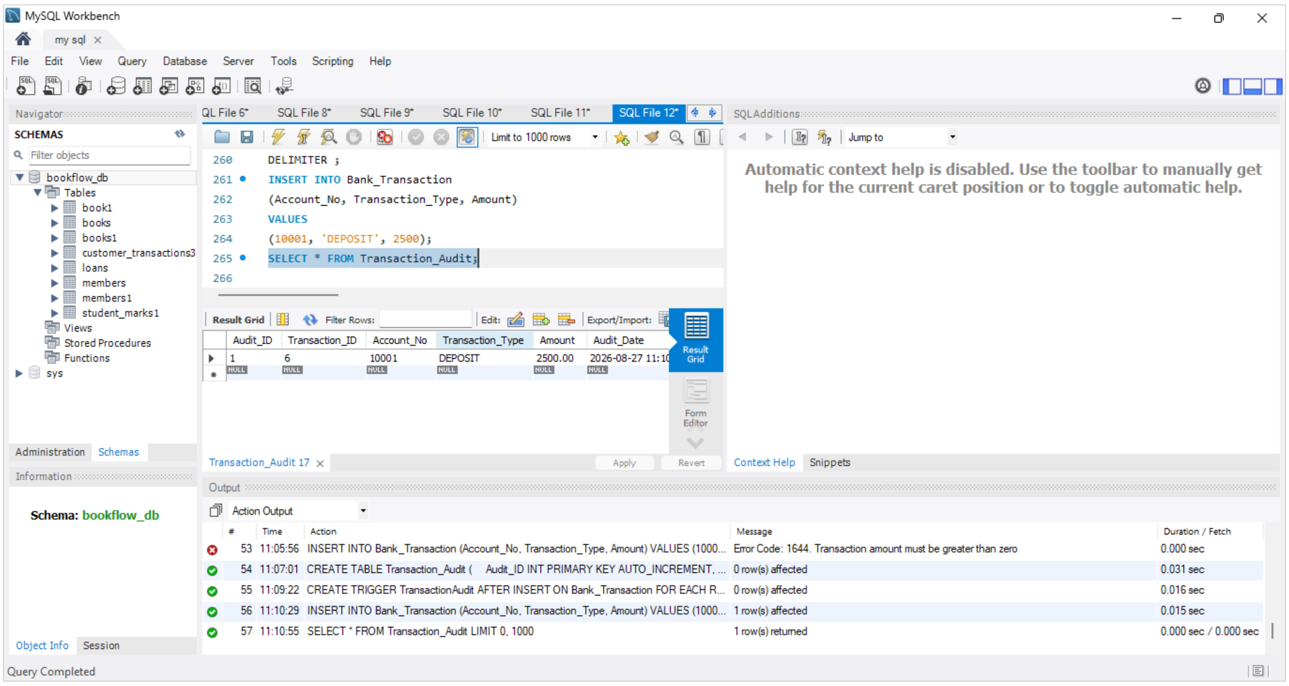
17. TRIGGER – PREVENT NEGATIVE DEPOSIT



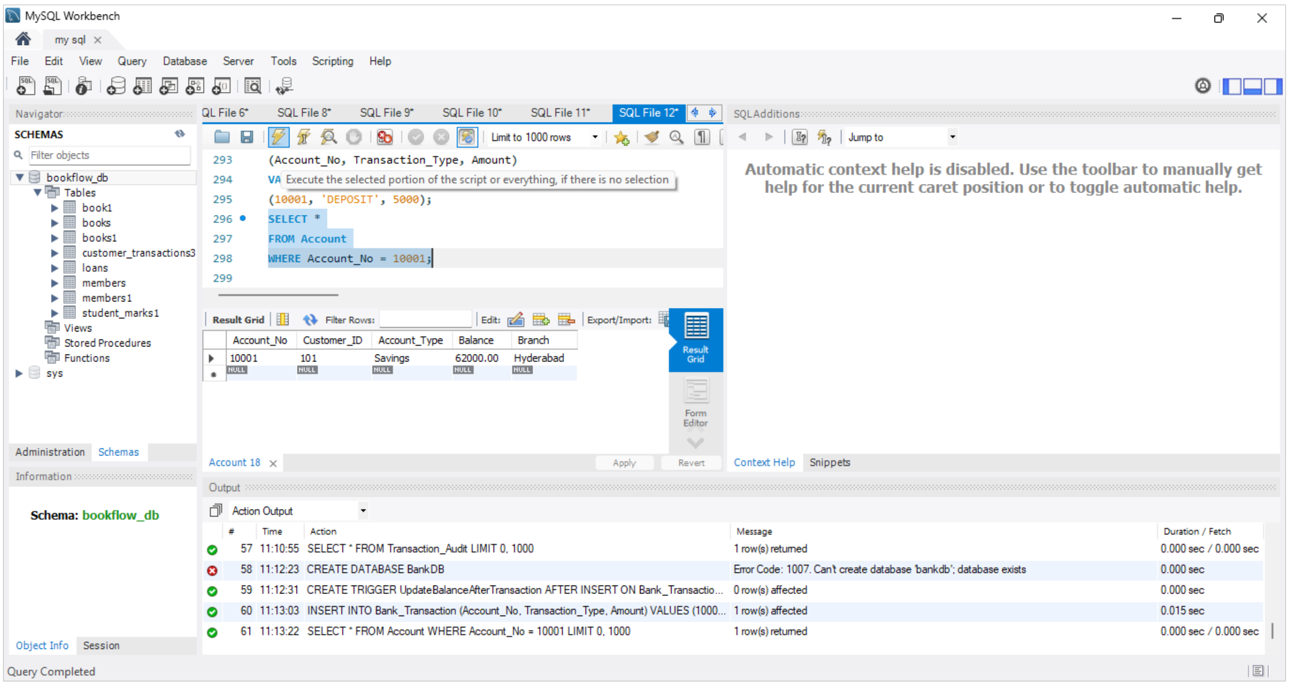
18. CREATE TRANSACTION AUDIT TABLE



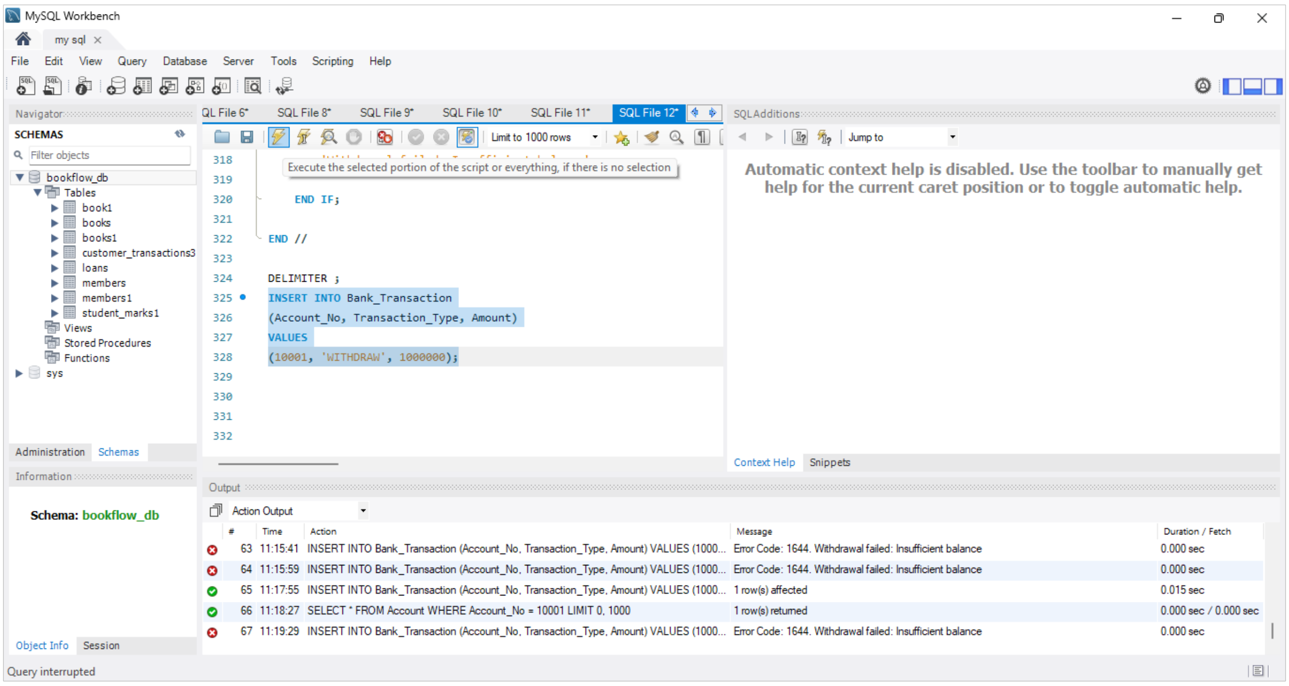
19. TRIGGER – TRANSACTION AUDIT



20. TRIGGER – AUTOMATICALLY UPDATE ACCOUNT BALANCE

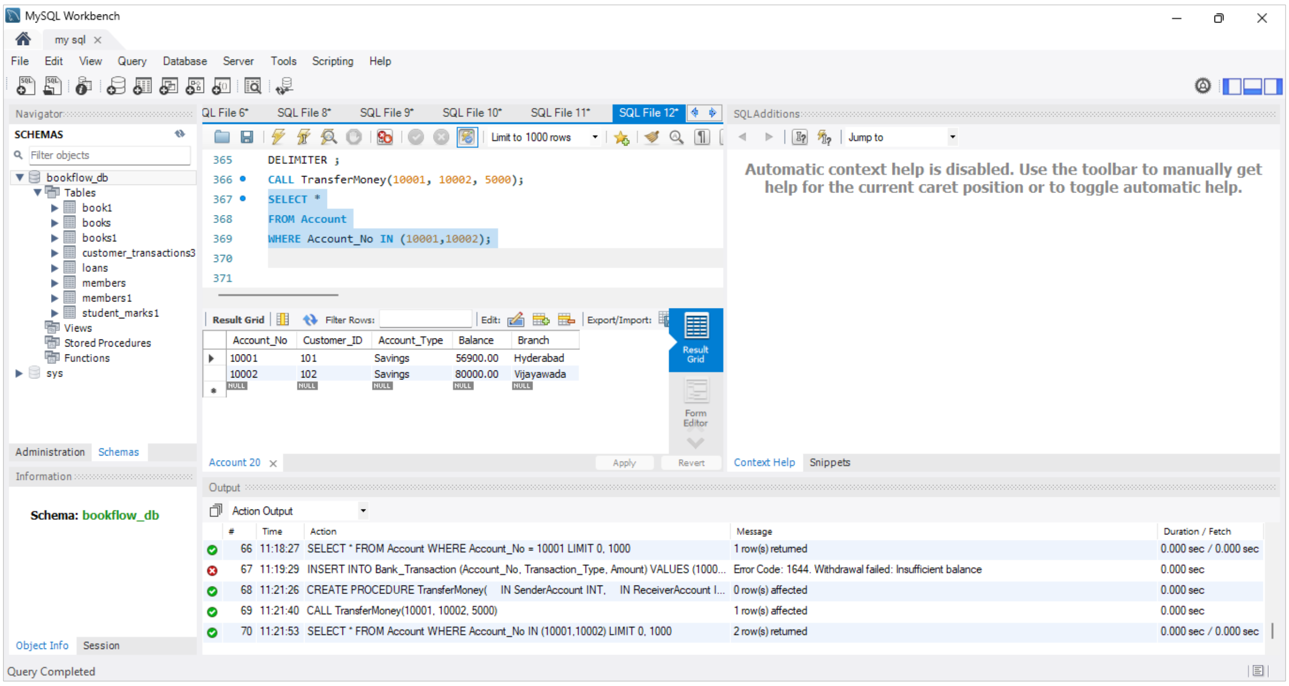


21. TRIGGER – PREVENT WITHDRAWAL ABOVE BALANCE

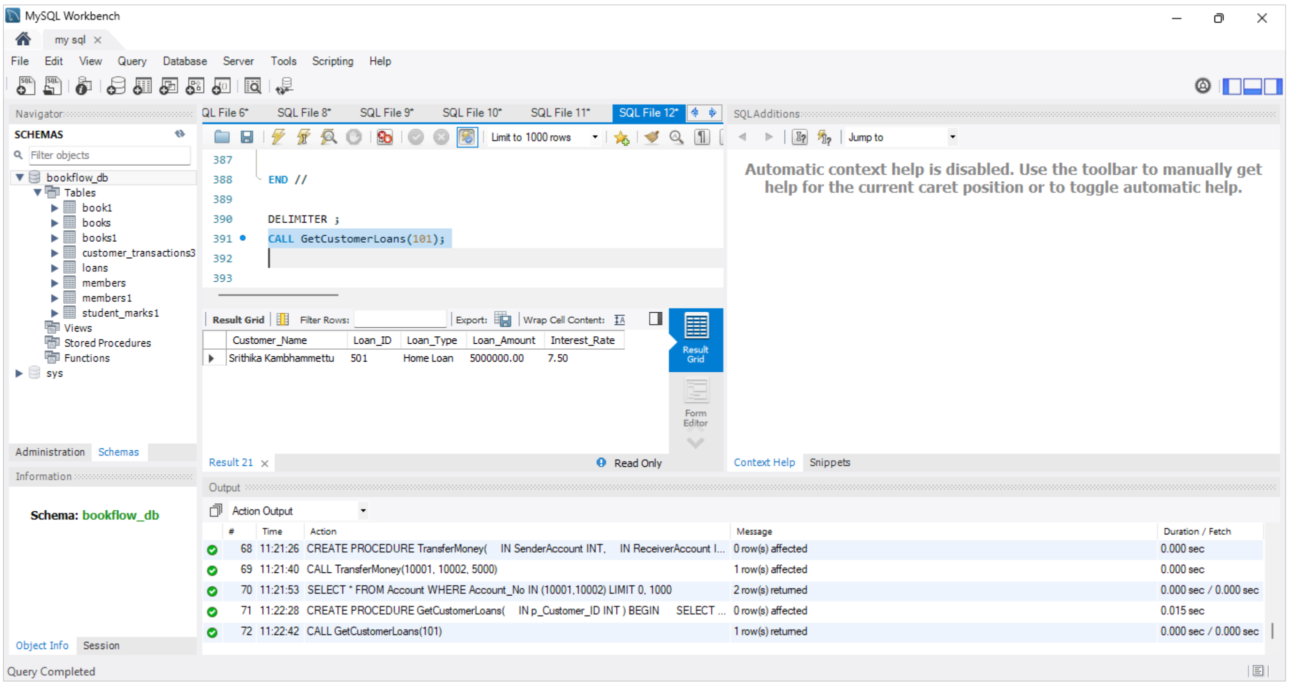


ADVANCED STORED PROCEDURES

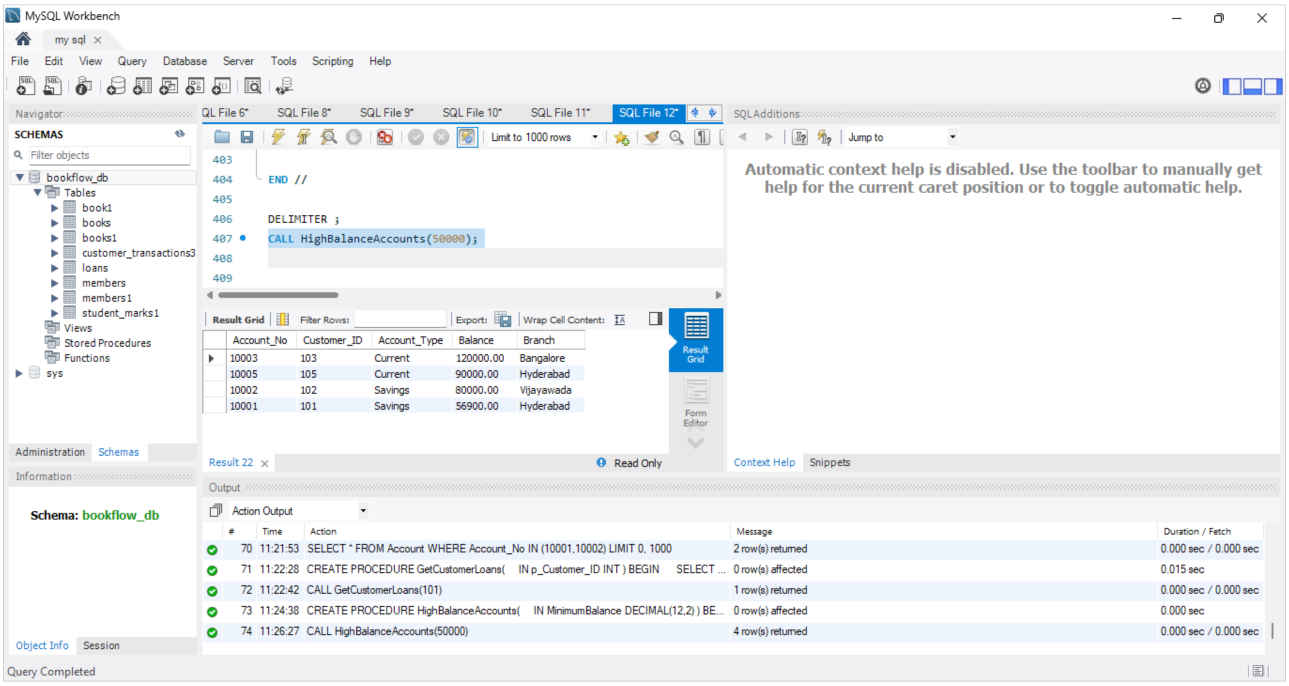
22. PROCEDURE – ACCOUNT TRANSFER



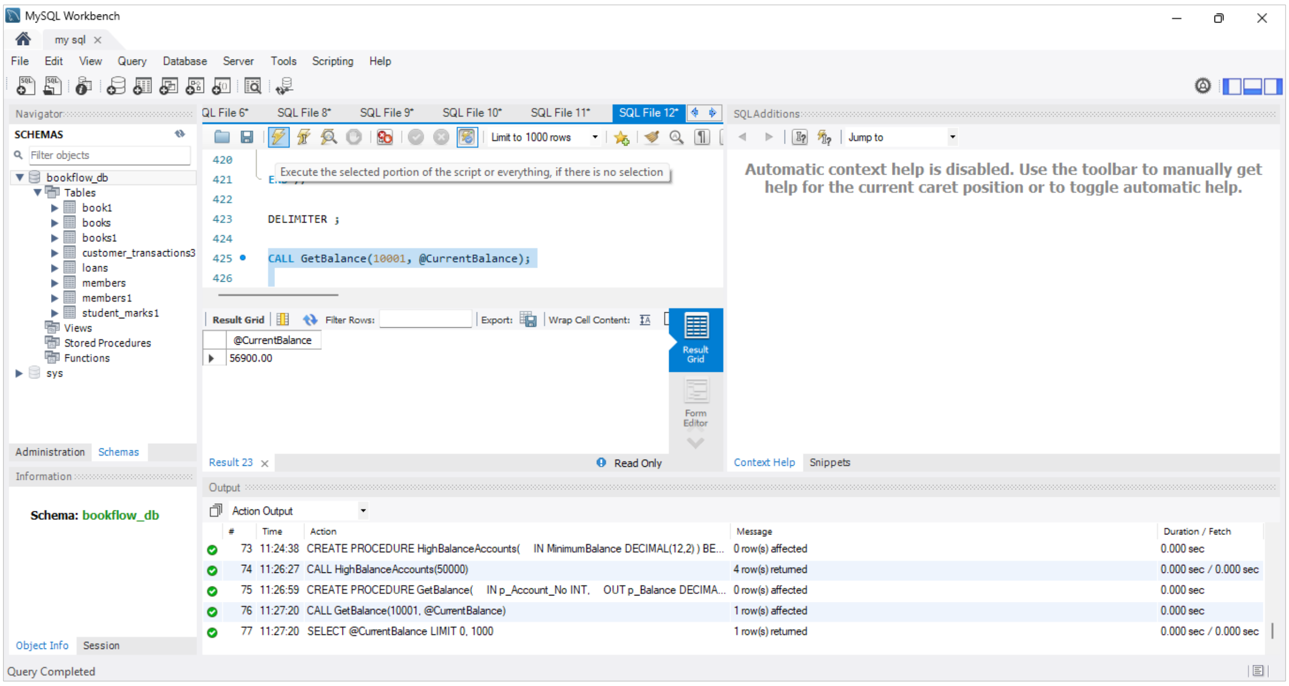
23. PROCEDURE – CUSTOMER LOAN DETAILS



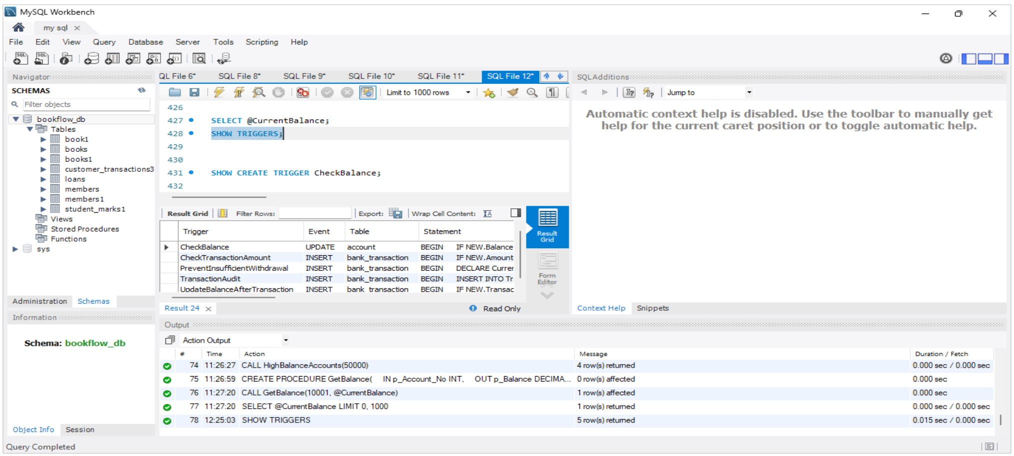
24. PROCEDURE – HIGH BALANCE ACCOUNTS

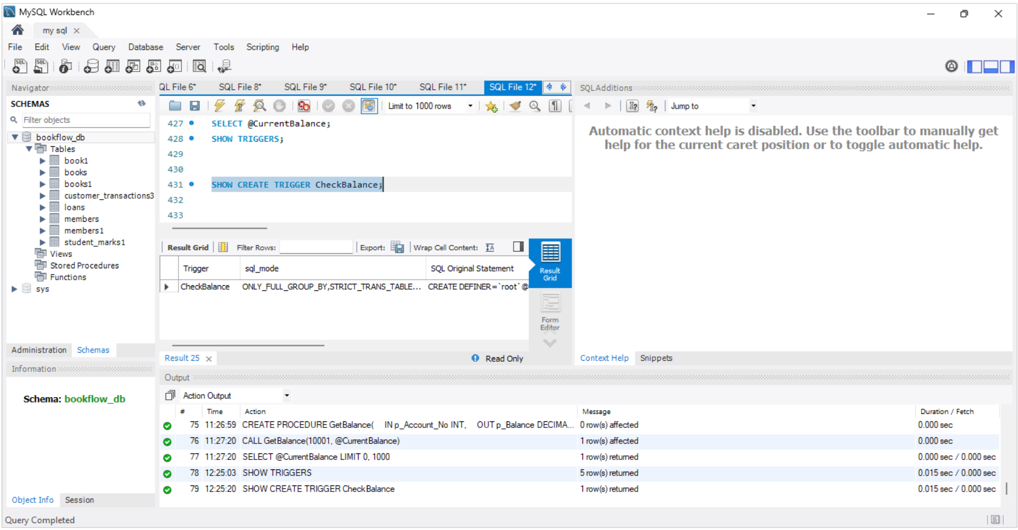


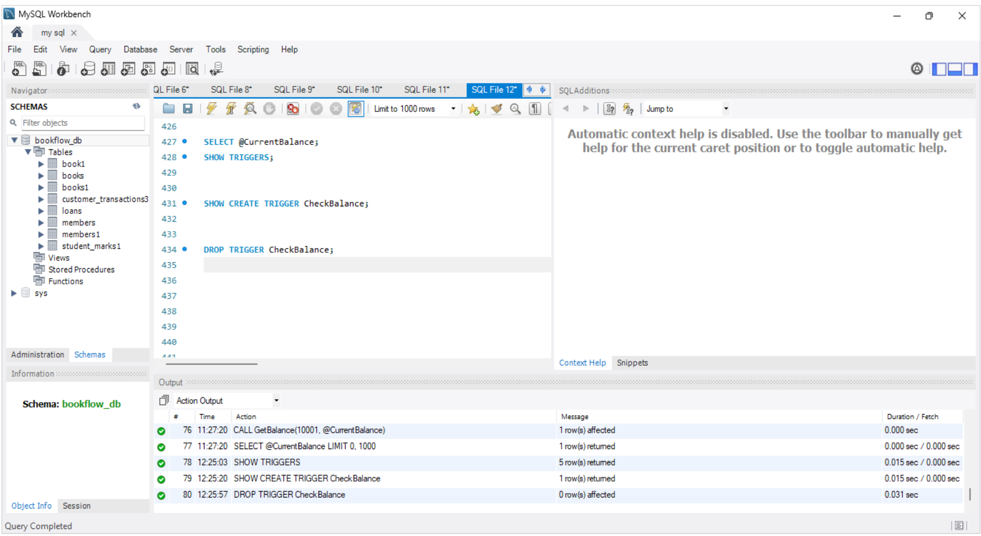
25. PROCEDURE WITH IN, OUT PARAMETERS



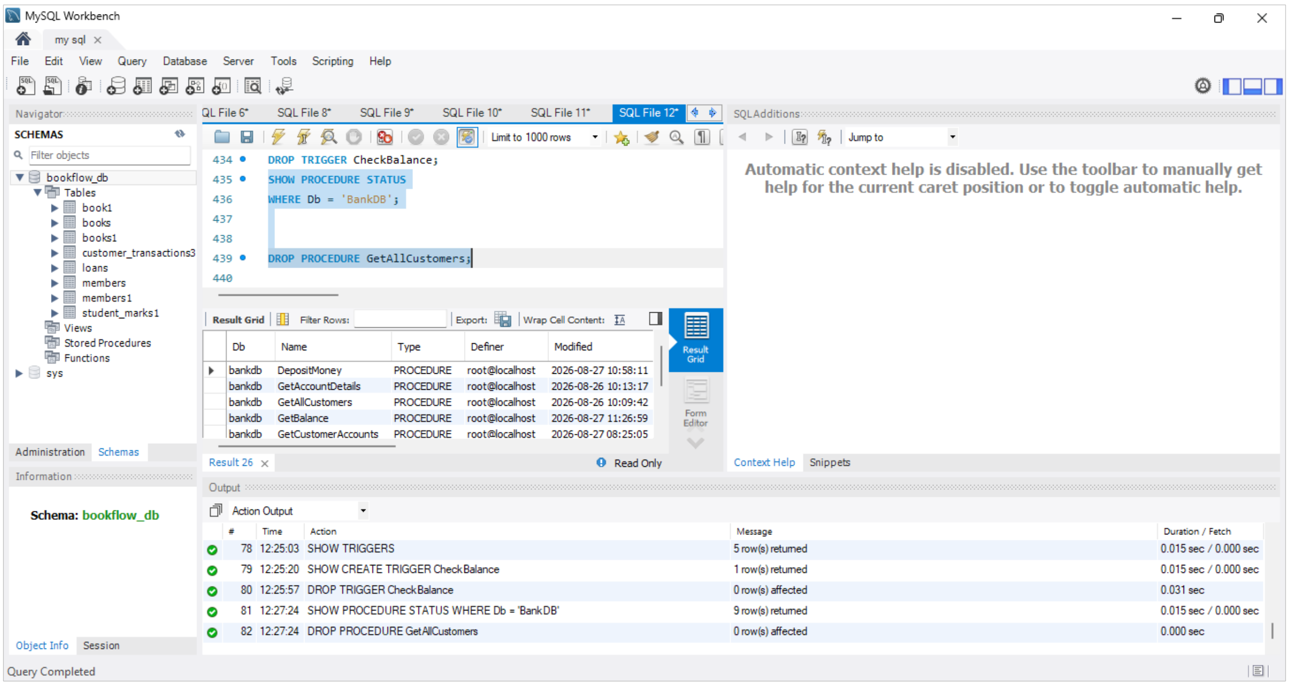
TRIGGER MANAGEMENT COMMANDS



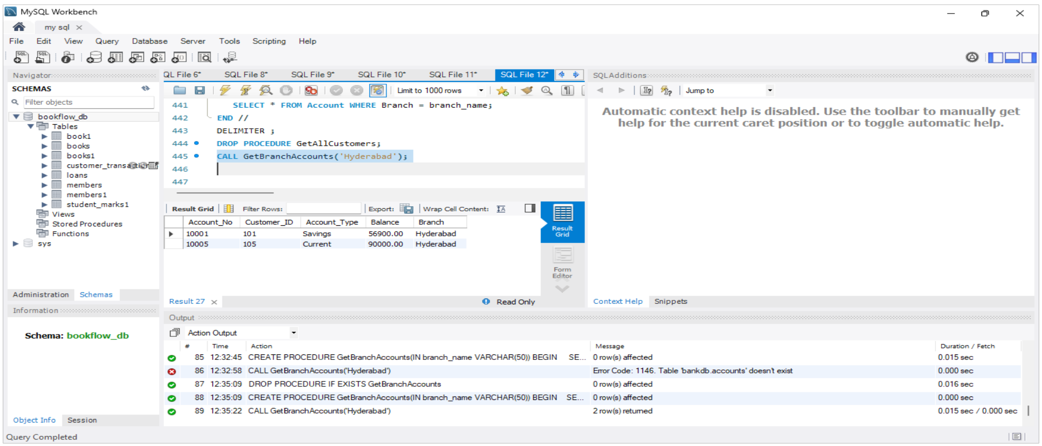




PROCEDURE MANAGEMENT COMMANDS

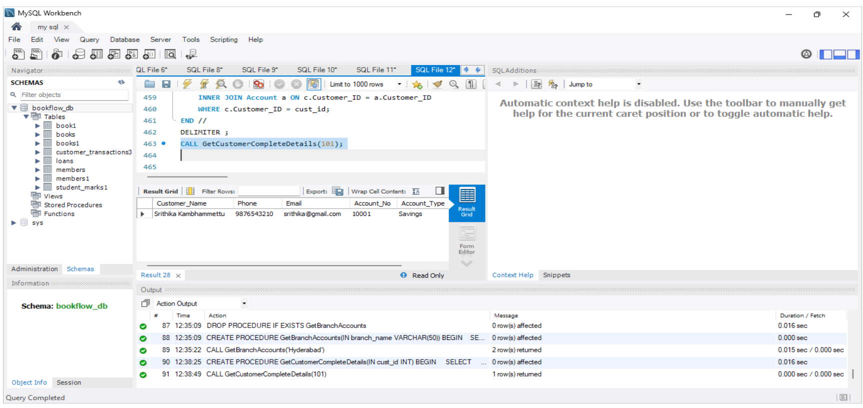


PRACTICE QUESTIONS



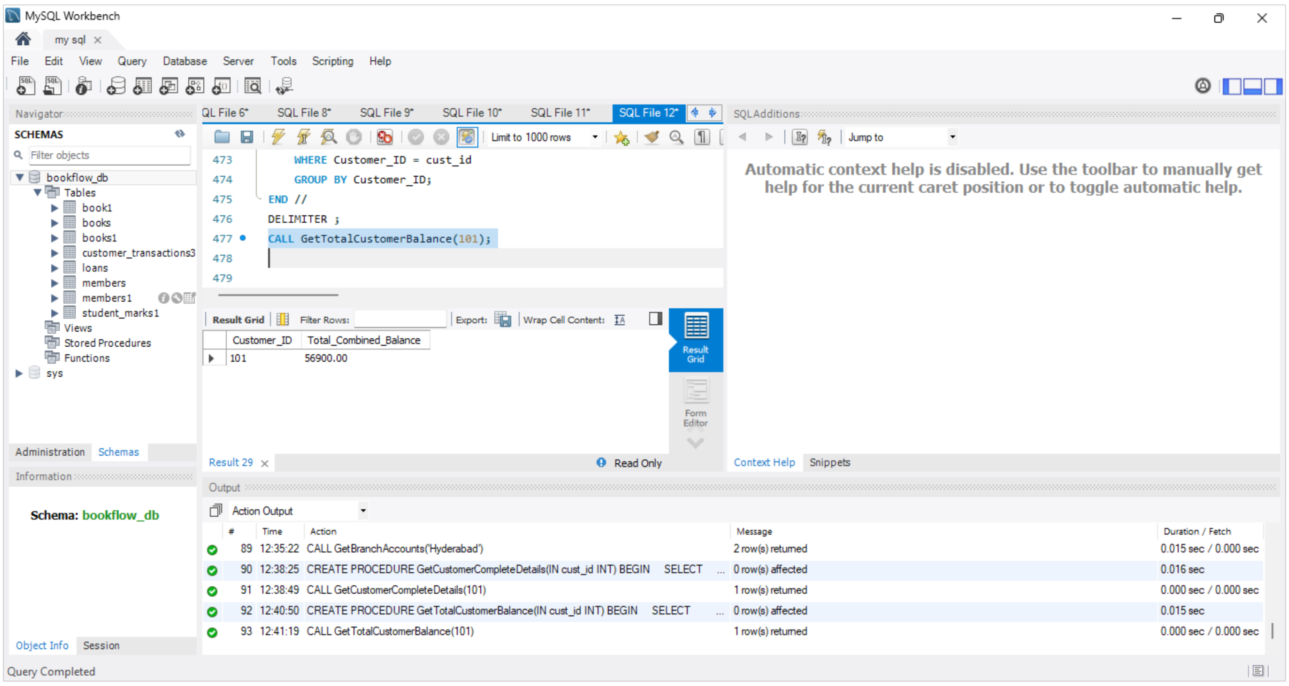
QUESTION 2:

Create a procedure that accepts Customer_ID and displays:



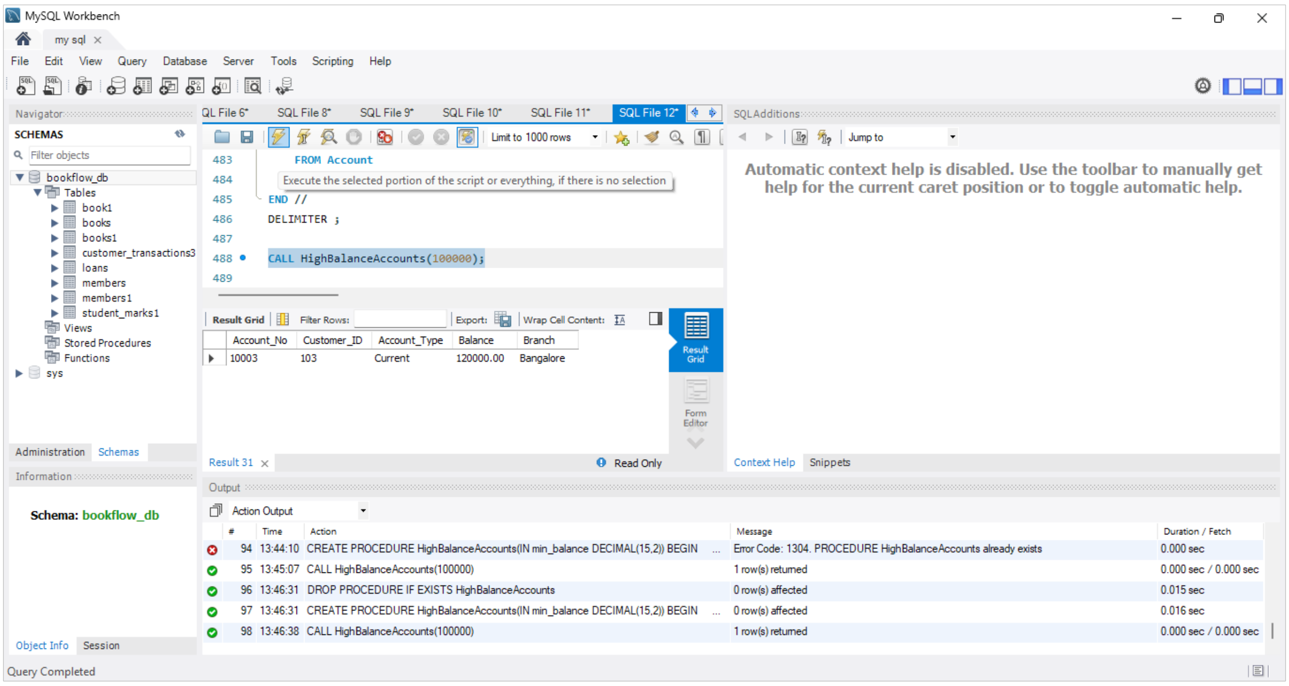
QUESTION 3:

Create a procedure to calculate the total balance of all accounts belonging to a particular customer.



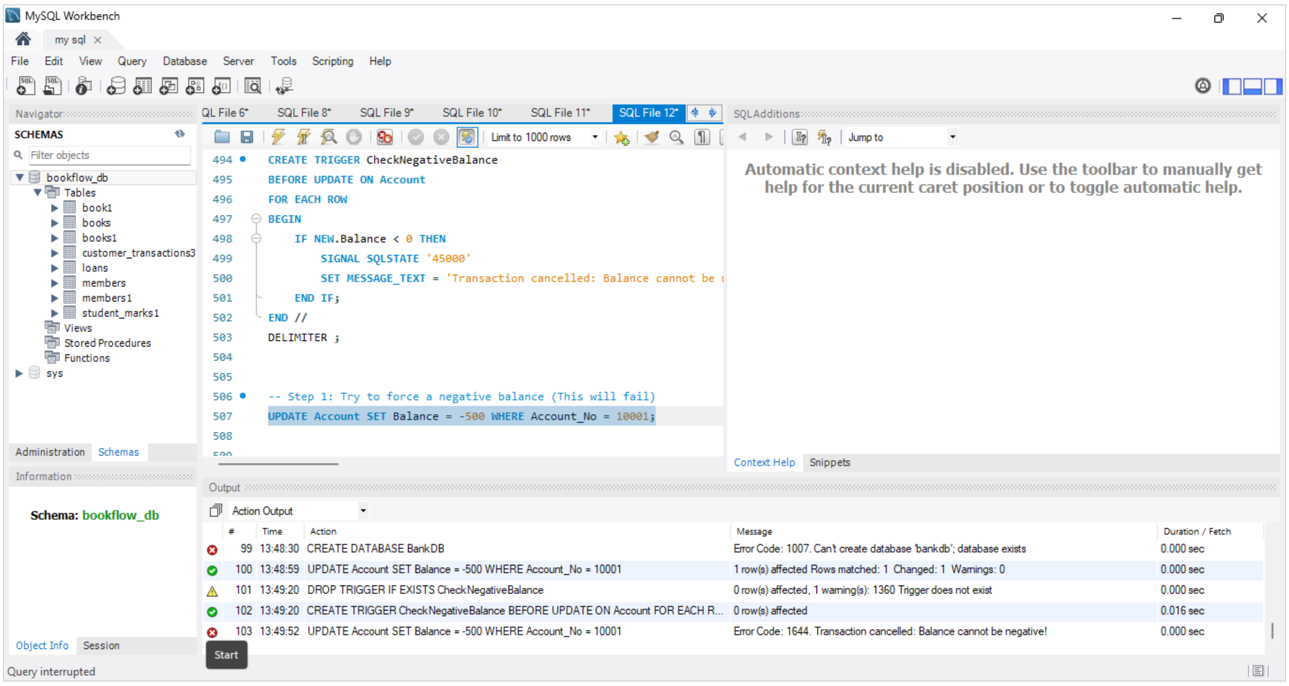
QUESTION 4:

Create a procedure to display accounts whose balance is greater than a given amount.



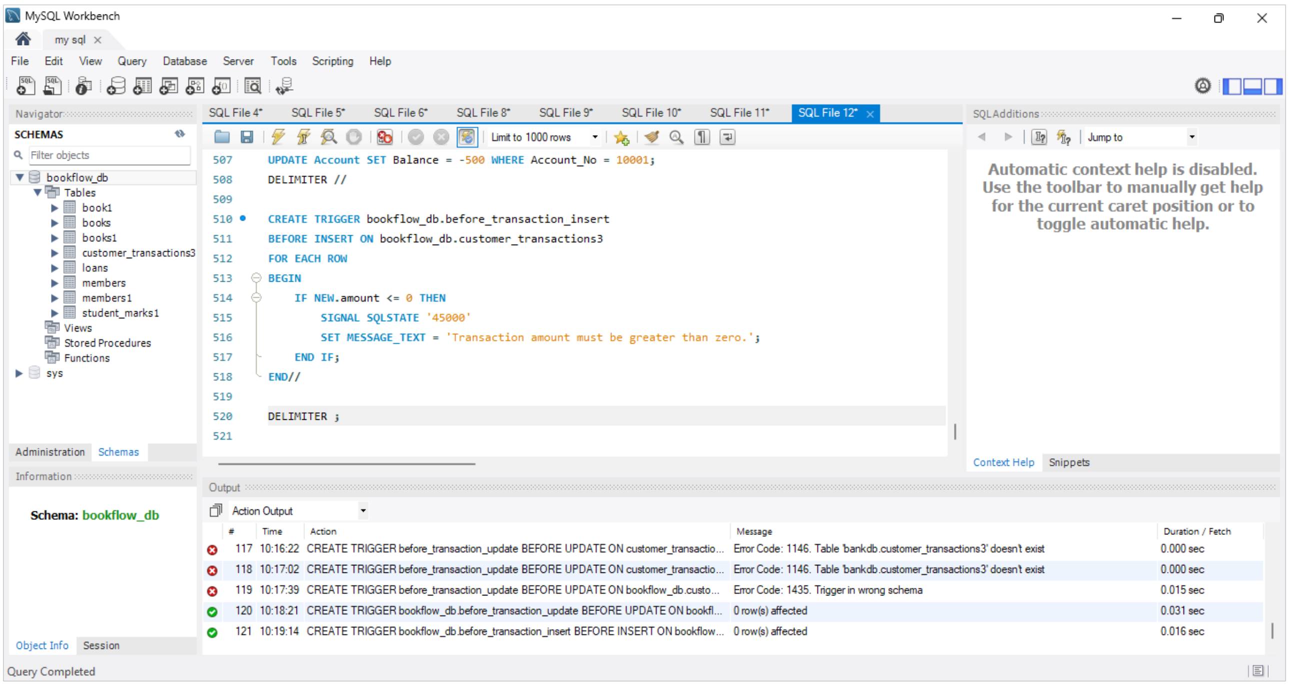
QUESTION 5:

Create a trigger that prevents an account balance from becoming negative.



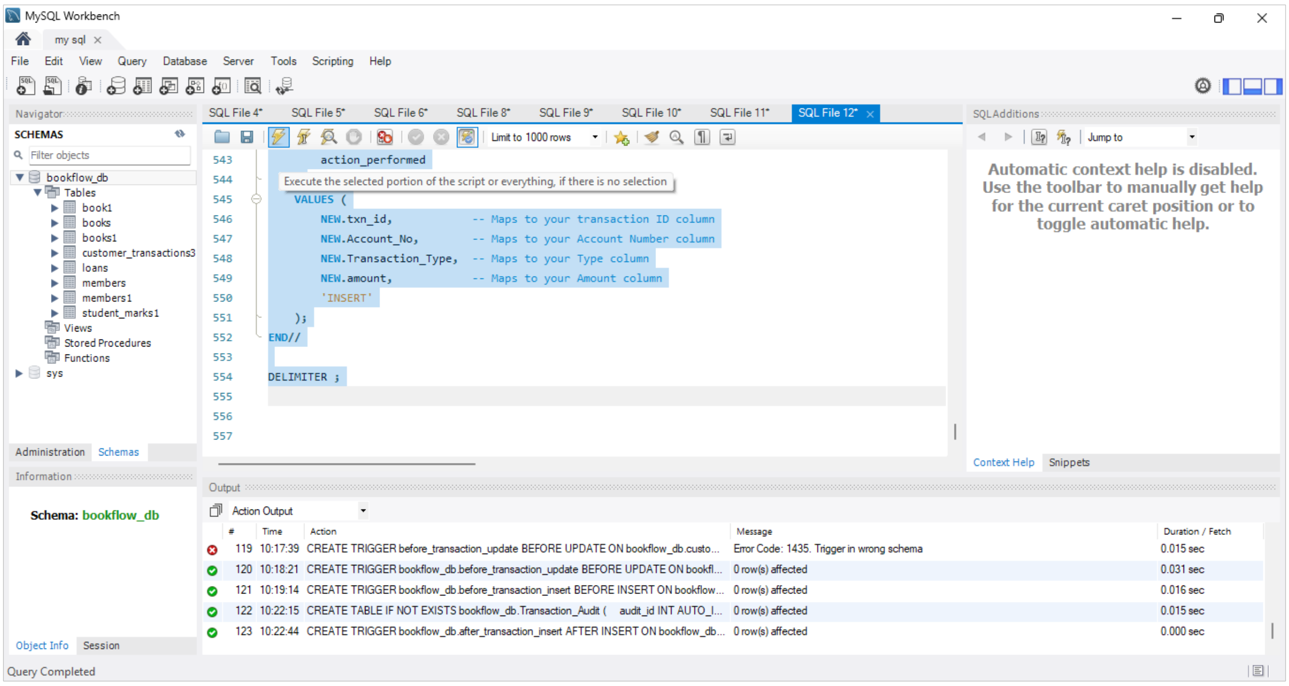
QUESTION 6:

Create a trigger that prevents a transaction amount from being zero or negative.



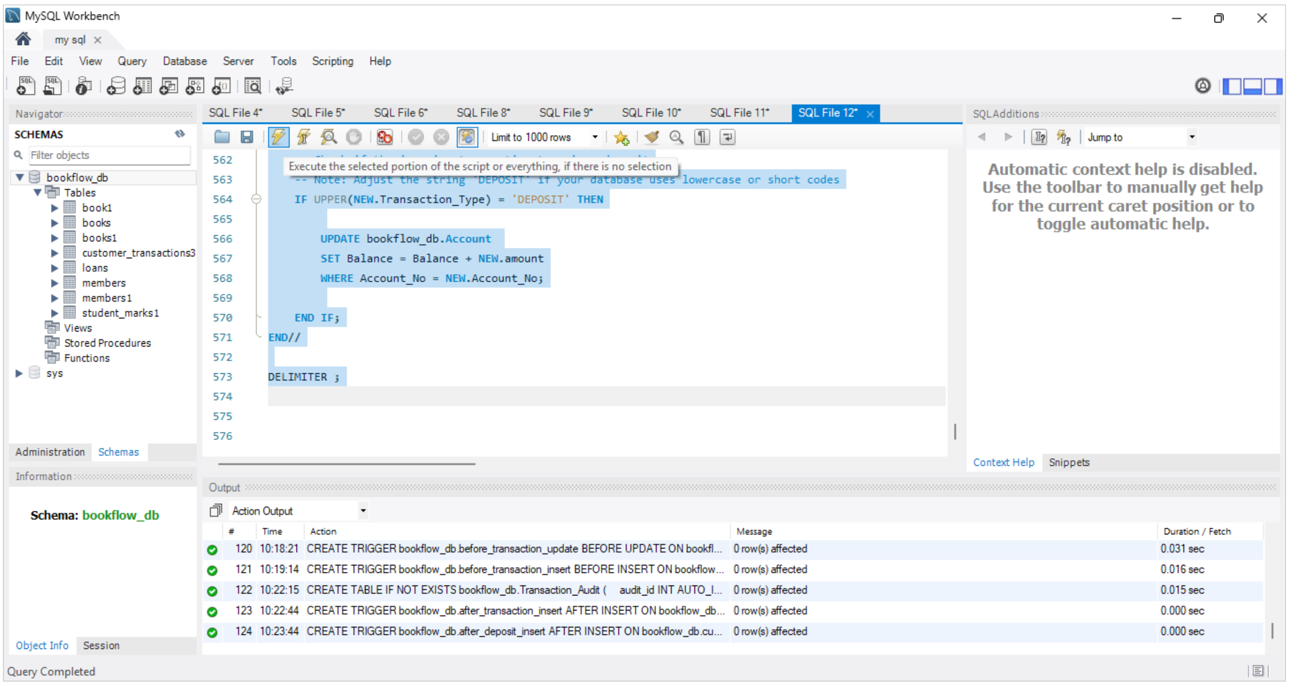
QUESTION 7:

Create a trigger that automatically records every transaction in Transaction_Audit.



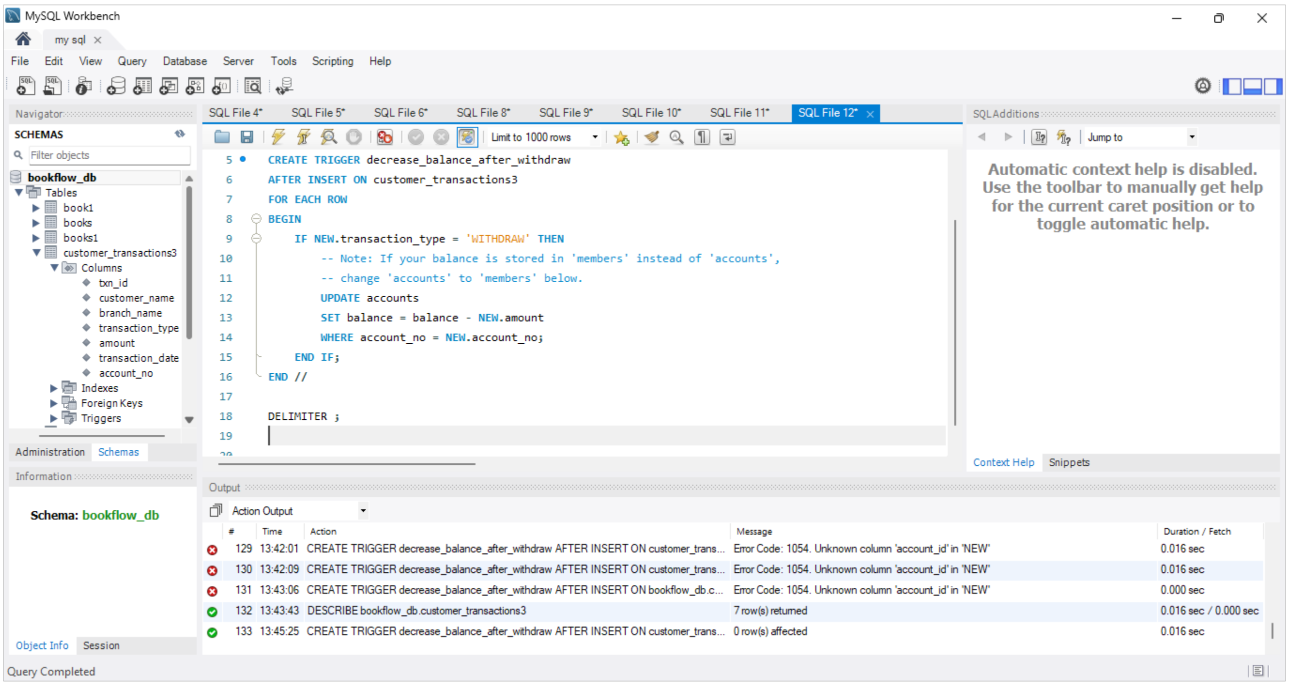
QUESTION 8:

Create a trigger that automatically increases the account balance whenever a DEPOSIT transaction is inserted.



QUESTION 9:

Create a trigger that automatically decreases the account balance whenever a WITHDRAW transaction is inserted.



QUESTION 10:

Create a trigger that prevents withdrawal when the account does not have sufficient balance.